PRISM-Net : A Pyramid Residual Integration and Separable Multi-scale U-Net for Polyp Segmentation.

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by

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TO

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Spring 26

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This thesis report is dedicated to us, our supervisor and our family. The team work was satisfactory and the family's support was incredibly Amazing. Our dedicated and most hard working Supervisor, who has been a constant support throughout these months. In this document, the contributions are acknowledged too.

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ETHICAL STATEMENT

We would like to declare that No unethical acts were committed to accomplish this thesis. The data, for the purpose of research, is authentic. Each one of the citations has been carefully verified. The authors of this thesis take on all responsibilities for any transgression of thesis rules.

Abstract

Colorectal polyp segmentation from endoscopic images is an important task in medical image analysis because accurate localization of polyp regions can support computer-aided gastrointestinal disease analysis and provide precise pixel-level information for subsequent clinical assessment. However, automatic polyp segmentation remains challenging due to considerable variations in polyp size, shape, texture, color, and appearance across endoscopic images. The segmentation task is further complicated by unclear polyp boundaries and visual similarity between the polyps and the surrounding mucosal tissues. In addition, polyps can appear at various scales, so it is challenging for standard segmentation models to simultaneously capture the local fine-grained features and overall context information. Existing encoder-decoder models can also generate irrelevant background features by skip connections, which may have negative effects on the segmentation performance especially in complex polyp borders. Therefore, an effective segmentation model is desired to learn hierarchical features with rich multi-scale contextual information, to selectively capture important polyp features and to reconstruct accurate segmentation boundaries. To address these challenges, we propose PRISM-Net (Pyramid Residual Integration and Separable Multi-scale U-Net for colon polyp segmentation), a U-Net based segmentation network, which employs a pretrained ResNet34 encoder, Atrous Spatial Pyramid Pooling (ASPP) module for feature extraction of multi-scale contextual features, attention gated skip connection for skip connection, and depth-wise separable convolution based decoder for efficient feature reconstruction. We propose to combine complementary feature extraction and feature selection mechanisms in a unified segmentation architecture. The performance of our proposed PRISM-Net are evaluated on the Kvasir-SEG dataset with various segmentation metrics including Dice coefficient, IoU, precision, recall, specificity, F1-score, and pixel accuracy. Experimental results demonstrate that PRISM-Net achieves a best validation Dice score of 0.9140 and a held-out test Dice score of 0.8873, with a test IoU of 0.8232 and F1-score of 0.9037. These results demonstrate that the proposed combination of residual feature extraction, attention-based feature selection, multi-scale contextual learning, and efficient feature reconstruction provides an effective approach for colorectal polyp segmentation.

Keywords: Colorectal Polyp Segmentation, Medical Image Analysis, Deep Learning, U-Net, ResNet34, Attention Mechanism, Multi-Scale Feature Learning, Kvasir-SEG

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Chapter 1
Introduction

1.1 RESEARCH GAP

Based on the reviewed approaches, several important observations can be made.

First, U-Net [24] provides effective spatial reconstruction through skip connections, but the original architecture does not explicitly incorporate residual feature extraction, attention-based feature selection, or multi-scale contextual learning.

Second, ResNet [23] provides strong residual feature extraction, but ResNet by itself is primarily an image recognition architecture and does not directly provide a complete pixel-level segmentation framework.

Third, attention-based approaches such as Attention U-Net [16] can improve the selection of relevant spatial features, but attention does not by itself solve the problem of representing objects at different spatial scales.

Fourth, multi-scale approaches such as ASPP [22] provide contextual information across different receptive fields, but they do not inherently provide attention-guided skip-feature selection.

Finally, computationally efficient convolutional operations can reduce decoder complexity, but efficiency alone does not guarantee accurate segmentation boundaries.

Therefore, this research addresses the gap by developing PRISM-UNet , which integrates:

Residual Feature Extraction + **Attention-Guided Feature Selection**
+ **Multi-Scale Contextual Learning** + **U-Net-Based Spatial Reconstruction**
+ **Computationally Efficient Decoding**

within a single segmentation model.

The research further investigates the contribution of these components through experimental evaluation rather than assuming that every architectural component automatically improves performance.

1.2 PROBLEM STATEMENT

Colorectal polyp segmentation from endoscopic images is a challenging computer vision problem because of the considerable visual variation present in polyp images. Polyps may have different sizes, shapes, colors, textures, and boundary characteristics. The neighboring mucosa also have a visual aspect similar to that of the polyp area such that it is hard to differentiate the foreground and the background.
The availability of annotated medical image dataset is the other difficulty. Kvasir-SEG has 1000 polyp images along with segmentation masks [21]. It can be used as a performance assessment of automatic polyp segmentation techniques. The relatively limited quantity of annotated images

implies a better feature representation should be used and the suitable training strategy should be adopted.
Moreover the nature of multiple scales about colorectal polyps is another difficulty. Segmentation model has to recognize both small and big polyps, while retaining its accuracy at boundaries. Conventional convolution operation, may fail to provide enough context information for objects at widely different scales. Using multi-scale feature extraction techniques such as ASPP [22], different dilation rates are used for extracting feature in different receptive fields.
The transmission of unwanted information through U-net's skip-connections is another trouble for model. Skip-connection preserves the spatial information, while only a small number of features produced by encoder are useful to segmentation model. Attention Gate model can perform such useful tasks. With the help of Attention Gate model, the useful spatial areas of segmentation model focus on irrelevant region is neglected.
It is another difficulty to restore high detailed spatial information during the decoder process by conventional convolution operations. Computation can be increased while using many convolutional operations, Depthwise-separable convolution separates the spatial filter from the channel mix process, which greatly saves computation.

In conclusion, segmentation model should be able to:

- extract strong hierarchical features;
- capture contextual information at multiple scales;
- Pay attention to salient polyp features;
- Keep spatial information for skip-connection; and
- Restore segmentation masks efficiently with less computation.

With the aim of facing those difficulties this paper presents MBLNet comparing with developed PRISM-UNET which combined pretrained ResNet34 encoder, attention-gated skip connections, ASPP-based multi-scale feature extraction and depthwise separable decoder.

1.3 MOTIVATION

The main goal of this research is to suggest a segmentation architecture, which would be able to identify the regions containing polyps in endoscopic images accurately and take into account the specifics of such images.
The MBLNet implementation provides an efficient U-Net architecture based on MobileNetV2-style inverted-residual depthwise-separable blocks. It employs channel expansion, intermediate activation, and residual connections but does not employ a pretrained ResNet encoder, ASPP, or attention mechanism. The MBLNet code specifically describes it as an inverted-residual depthwise separable U-Net without attention or ASPP. Therefore, it provides a meaningful baseline for comparison.

Several complementing enhancements are introduced by the proposed PRISM-UNet:

- ResNet34 encoder to provide richer pretrained features;
- ASPP to achieve multi-scale context information;

- Attention Gates to selectively gather encoder information;
- Depthwise-separable decoder to build semantic masks.

The specific motivations behind the individual changes are outlined below:

1. **Polyp Variation**

   Polyp color,shape,size,texture, and visual appearance could vary greatly.So segmentation algorithm should be equipped to learn effective generalization features across polyp varied appearance.

2. **Rare annotated medical data**

   The collection of medical images requires annotations from medical experts, which is expensive and time-consuming. Provided 1,000 annotated polyp images were provided by Kvasir-SEG [21], it is necessary to extract effective representative features.

3. **Requirements for strong feature extractor**

   A pretrained ResNet34 encoder effectively provides a hierarchically structured visual representation, as well as capability of residual learning. Especially for deep architecture optimization, ResNet [23] has addressed it through residual connection.

4. **Requirements for multi-scale feature extractor**

   A polyp might take place in various scales in the endoscopic image. Thus we integrate ASPP module, so as to capture context information using receptive fields in multiple sizes.

5. **Requirements for selective feature fusion**

   Endoscopic images present significant amount of irrelevant background content. Attention mechanism could be beneficial for the segmentation network by forcing it to focus on necessary features for polyp identification while reducing features related to irrelevant background.

6. **Requirements for efficient feature reconstructions**

   High-resolution segmented mask needs to be reconstructed at reasonable efficiency. Hence depthwise-separable convolution is included for feature reconstruction in decoder, which achieves an efficient reconstruction process.

   These motivations collectively led to the development of PRISM-UNet for colorectal polyp segmentation.

1.4 OBJECTIVE OF RESEARCH

The overall aim of this research is to develop and evaluate an effective deep learning-based model for colorectal polyp segmentation from endoscopic images.
The research has three specific objectives:

1. To develop PRISM-UNet (Pyramid Residual Integration and Separable Multi-scale U-Net for colon polyp segmentation) by integrating a pretrained ResNet34 encoder, an ASPP-based multi-scale feature extraction module, attention-gated skip connections, and a depthwise-separable convolution-based decoder within a U-Net architecture.

2. To design and analyze the major modules of PRISM-UNet, including the ASPP multi-scale module, attention-gated skip connections, and depthwise-separable decoder, and investigate their contributions to the overall segmentation performance.

3. To conduct comprehensive experimental evaluation of the proposed model's performance and robustness, including multi-seed statistical validation with 95% confidence intervals and zero-shot cross-dataset generalization testing on CVC-ClinicDB, CVC-ColonDB, and ETIS-LaribPolypDB.

1.5 ORGANIZATIONAL STTRUCTURE OF THIS THESIS

This thesis consists of five chapters.

Chapter I: Introduction presents the research background, problem statement, motivation, research objectives, and organization of the thesis.

Chapter II: Literature Review discusses the scope of the research and reviews existing research related to biomedical image segmentation, U-Net, residual learning, attention mechanisms, multi-scale feature extraction, efficient convolution, and colorectal polyp segmentation.

Chapter III: Methodology presents the research methodology, dataset, preprocessing procedures, MBLNet baseline, proposed PRISM-UNet architecture, individual modules, training strategy, loss function, evaluation metrics, and experimental setup.

Chapter IV: Results and Discussion presents and analyzes the experimental results obtained from RAMS-UNet. This chapter includes quantitative evaluation, module-level experiments, comparative experiments, and qualitative segmentation analysis.

Chapter V: Conclusion and Future Works summarizes the research findings, major contributions, limitations, and possible directions for future research.

Chapter 2
Literature Review

2.1 INTRODUCTION

Deep learning has become an important approach for medical image segmentation because deep neural networks can automatically learn hierarchical representations from images. In segmentation tasks, the objective is to assign a class label to individual pixels so that a target anatomical structure, lesion, or pathological region can be separated from the surrounding background.
The development of biomedical segmentation architectures has progressed from conventional convolutional networks toward encoder-decoder architectures, residual networks, attention-based models, multi-scale feature extraction approaches, and computationally efficient convolutional architectures.
For colorectal polyp segmentation, these developments are particularly relevant because endoscopic images contain considerable variation in polyp appearance and scale. The Kvasir-SEG dataset provides 1,000 polyp images and corresponding segmentation masks and has been used as a benchmark for evaluating automatic polyp segmentation methods [21].
This chapter reviews the techniques directly related to the two architectures investigated in this research: MBLNet as the comparison model and PRISM-UNet as the proposed model. The review focuses on U-Net, residual feature extraction, attention mechanisms, multi-scale feature learning, depthwise-separable convolution, and deep learning-based colorectal polyp segmentation.

2.2 SCOPE OF RESEARCH

The scope of this research focuses on deep learning-based semantic segmentation of gastrointestinal polyps from endoscopic images.

The research specifically considers:

1. **Medical image segmentation:** Pixel-level identification of polyp regions.
2. **CNN-based segmentation:** Encoder-decoder architectures for extracting and reconstructing image features.
3. **U-Net architecture:** The baseline segmentation framework used as the foundation of the proposed model.
4. **ResNet34 encoder:** A pretrained residual network used to improve hierarchical feature extraction.
5. **ASPP module:** Used to capture multi-scale contextual information.
6. **Attention gates:** Used to suppress irrelevant features in skip connections.
7. **Depthwise-separable decoder:** Used to construct a more computationally efficient decoder.

8. **Kvasir-SEG dataset:** Used for training and evaluating the segmentation models.
9. **Ablation analysis:** Used to examine the individual contribution of the architectural components.

The primary target is the segmentation of the polyp region and background region in endoscopic images.
The research does not focus on histopathological classification, cancer grading, bounding-box object detection, or clinical diagnosis. The primary research task is pixel-level colorectal polyp segmentation.

2.3 LITERATURE REVIEW

2.3.1 Overview

Automated segmentation of colorectal polyps from colonoscopy images has become one of the most actively studied problems in medical image analysis, since accurate delineation of the polyp region directly supports early diagnosis and reduces the risk of interval colorectal cancer. Building on the encoder-decoder paradigm introduced by U-Net, recent research has pursued two, sometimes competing, goals: (i) pushing segmentation accuracy higher through richer attention mechanisms, multi-scale context modules, and transformer/foundation-model backbones, and (ii) keeping the resulting models light enough for point-of-care (PoC) and real-time clinical deployment. This chapter reviews recent literature (primarily 2017-2026) on deep learning-based polyp segmentation, with particular emphasis on lightweight and attention-augmented U-Net variants, ResNet-style encoders, ASPP-based multi-scale context modules, and depthwise-separable design choices the same design space that the proposed model (a pretrained ResNet-34 encoder combined with an ASPP bottleneck, attention-gated skip connections, and a depthwise separable decoder) occupies.

2.3.2 Literature Review based on Deep Learning and CNN for Polyp Segmentation

Recent advances in deep learning have substantially improved automatic colorectal polyp segmentation. However, the development of accurate models must also consider computational complexity, boundary delineation, generalization across datasets, and suitability for real-time clinical applications. The following papers highlight some illustrative progresses over 2017-2026, and specifically, the encoder-decoder structures, attention mechanisms, multi-scale feature extraction, lightweight convolution, and pre-trained features representations.

2.3.2.1 Study 1: BAR-UNet for Boundary-Aware and Appearance-Robust Polyp Segmentation

In [1], Xu et al. proposed BAR-UNet, a U-Net-style architecture employing a pretrained ResNet-34 encoder and a decoder designed specifically for colorectal polyp segmentation. The architecture incorporates a boundary-aware head to explicitly learn polyp boundaries and a mask-guided appearance consistency learning (MACL) strategy to improve robustness against variations in endoscopic appearance. The use of a ResNet-34 encoder is particularly relevant to the proposed study because it demonstrates the effectiveness of a pretrained ResNet-based feature extractor within a U-Net framework.

The model was evaluated on Kvasir-SEG and further tested on CVC-ClinicDB, CVC-ColonDB, and ETIS-Larib for cross-dataset generalization. On Kvasir-SEG, BAR-UNet achieved a Dice score of 0.8798 $\pm$ 0.0024 and an IoU of 0.7913 $\pm$ 0.0029. In zero-shot evaluation, it achieved Dice/IoU values of 0.831/0.748 on CVC-ClinicDB, 0.776/0.682 on CVC-ColonDB, and 0.728/0.631 on ETIS-Larib. These results demonstrate strong boundary precision and cross-dataset robustness. However, the model introduces additional boundary and consistency-learning components and is therefore more complex than a conventional lightweight U-Net. Prospective clinical validation is also still required [1].

2.3.2.2 Study 2: Polyp-DiFoM for Foundation-Model Knowledge Distillation

Agnihotri et al. introduced Polyp-DiFoM, a framework that transfers knowledge from large foundation models into lightweight polyp segmentation networks [2]. Instead of directly deploying computationally expensive foundation models, the method distills complementary information from models such as SAM and DINOv2 into lightweight U-Net and U-Net++ style baselines. This approach combines semantic and low-level feature information to improve the representation capability of compact segmentation models.
The framework was evaluated on Kvasir-SEG, CVC-ClinicDB, CVC-300, CVC-ColonDB, and ETIS datasets. The experiments demonstrated that foundation-model knowledge can improve lightweight baselines and provide competitive segmentation performance while retaining a substantially smaller inference architecture than the original foundation models. However, the approach requires large pretrained teacher models during the training stage, which increases training complexity and resource requirements. Therefore, although the final model can remain lightweight, the overall training pipeline is not as simple as conventional end-to-end CNN training [2].

2.3.2.3 Study 3: UNeSt A Lightweight MLP and Deep Separable Convolution Network

Li et al. proposed UNeSt, a lightweight colorectal polyp segmentation network that combines MLP-based feature modeling, deep/depthwise-separable convolution, and attention mechanisms [3]. The model was designed primarily for accurate segmentation under low computational and parameter budgets, making it highly relevant to point-of-care applications. Its architecture employs UNeXt as the backbone and introduces depthwise-separable convolution together with CBAM and Coordinate Attention modules.
The authors trained the model using Kvasir and CVC-ClinicDB and evaluated its generalization on CVC-300, ETIS, and CVC-ColonDB. UNeSt achieved 0.92M parameters and 0.23 GFLOPs with an inference time of 0.5249 s. It obtained F1/IoU values of 86.50/79.09 on CVC-ClinicDB and 82.69/72.85 on Kvasir. On the external datasets, it achieved 79.42/67.09 on CVC-300, 55.12/43.09 on ETIS, and 64.06/51.35 on CVC-ColonDB. Compared with the UNeXt backbone, UNeSt improved the CVC-ClinicDB F1 score from 82.81 to 86.50 while reducing the parameter count from 1.47M to 0.92M. The ablation study therefore demonstrated that the combination of deep separable convolution and attention modules can improve the accuracy–efficiency trade-off [3].
Despite its strong efficiency, UNeSt still showed substantially lower performance on challenging external datasets such as ETIS and CVC-ColonDB. In particular, the ETIS F1 score was only 55.12, indicating that ambiguous boundaries and domain differences remain difficult. Furthermore, the model is based on an MLP/UNeXt-style backbone rather than a pretrained ResNet encoder, leaving room to investigate whether pretrained residual features combined

with a lightweight decoder can improve generalization. The experimental results and ablation findings are reported in the Base Paper [3].

2.3.2.4 Study 4: Lightweight Deep Learning for Polyp Segmentation

Recent reviews of colorectal polyp segmentation have highlighted the transition from conventional U-Net models toward attention-enhanced CNNs, lightweight architectures, transformer-based models, and foundation-model-based approaches [4]. The reviewed methods show that U-Net-based architectures remain attractive because of their simple encoder–decoder structure and effective skip connections, whereas attention and transformer mechanisms can improve contextual and boundary representation.
Across the reviewed studies, however, higher segmentation accuracy is frequently associated with increased parameter count and computational complexity. Lightweight CNN architectures therefore remain important for real-time or point-of-care applications. The review also indicates that small and difficult datasets such as ETIS and CVC-ColonDB remain challenging because of low contrast, ambiguous boundaries, and domain variation. A major limitation identified in this line of research is the difficulty of simultaneously achieving high segmentation accuracy, low computational cost, and strong cross-dataset generalization [4].

2.3.2.5 Study 5: Li-DeepLabV3+ for Lightweight Multi-Scale Polyp Segmentation

Xiang et al. proposed Li-DeepLabV3+, a lightweight colon polyp segmentation network based on an improved DeepLabV3+ architecture [5]. The model uses an optimized MobileNetV2 backbone and lightweight multi-scale feature extraction, together with an attention-based feature fusion mechanism. This design is relevant to the proposed architecture because it demonstrates the usefulness of MobileNetV2-style lightweight feature extraction, multi-scale contextual information, and attention-based fusion for polyp segmentation.
The model was evaluated on Kvasir-SEG and CVC-ClinicDB, including cross-dataset experiments. On Kvasir-SEG, Li-DeepLabV3+ achieved an IoU of 79.88%, Dice of 88.82%, precision of 88.42%, recall of 89.22%, and accuracy of 96.56%. On CVC-ClinicDB, it achieved an IoU of 87.42% and Dice of 93.28%. In cross-dataset testing, training on Kvasir-SEG and testing on CVC-ClinicDB produced an IoU of 70.95% and Dice of 83.00%, while training on CVC-ClinicDB and testing on Kvasir-SEG produced an IoU of 68.58% and Dice of 81.36% [5]. Although Li-DeepLabV3+ provides a strong balance between accuracy and efficiency, its architecture relies on a specialized DeepLabV3+ design and MobileNetV2 backbone. The cross-dataset results also demonstrate that domain shifts can considerably reduce performance. Therefore, further investigation of pretrained residual encoders and more selective skip-feature fusion remains valuable.

2.3.2.6 Study 6: DeepNeXt–Lightweight Multi-Scale Attention Network

Wang et al. proposed DeepNeXt, a lightweight polyp segmentation model that integrates depthwise-separable convolution with a multi-scale attention feature encoding module [6]. The architecture is specifically designed for computationally constrained medical devices. Multi-scale attention is used to capture features at different spatial scales, while lightweight convolutional operations reduce the computational burden.
DeepNeXt was evaluated on Kvasir-SEG and CVC-ClinicDB. It contains approximately 1.51M parameters and requires only 3.04 GFLOPs. On Kvasir-SEG, DeepNeXt achieved an mIoU of 83.91%, Dice of 90.89%, precision of 93.35%, and recall of 91.85%. On CVC-ClinicDB, it

achieved an mIoU of 87.37%. These results demonstrate a strong balance between computational efficiency and segmentation accuracy [6].
However, although DeepNeXt is lightweight, it introduces multiple attention and multi-scale operations. Its primary evaluation focuses on Kvasir-SEG and CVC-ClinicDB, while more difficult external datasets can still expose domain-generalization limitations. This motivates architectures that can combine efficient decoding with stronger pretrained semantic representations.

2.3.2.7 Study 7: Implicit Edge-Guided Cross-Layer Feature Fusion

Liu et al. Presented an implicit edge-guided cross layer fusion approach to enhance segmentation of weak or fuzzy polyp boundary [7]. Different to skip connections between the corresponding same-level encoder and decoder layer, information is dispersed across all encoder and decoder layers without simple same-level connection. The cross layer feature fusion aims at integrating the low-level spacial knowledge with high-level semantics.
The model was assessed on typical colorectal polyp segmentation benchmarks and showed promising results in boundary description in comparison to standard skip-connection fusion approaches. The experiments have demonstrated that the explicit enrichment of boundary information can lead to a decrease in contour errors in terms of challenging colonoscopy images. Additional feature-fusion operations, nevertheless, result in complex architectures and it fails to fully exploit the efficiency of depth-wise separable decoder blocks.

2.3.2.8 Study 8: MEGANet for Multi-Scale Edge-Guided Attention

Bui et al. proposed MEGANet, a multi-scale edge-guided attention network designed to improve segmentation of polyps with weak or low-contrast boundaries [8]. The proposed architecture integrates multi-scale feature extraction and edge-guided attention mechanism, so as to focus on the salient region and border of polyp.
On Kvasir-SEG, MEGANet based on Res2Net-50 achieved an mDice of approximately 0.913 and mIoU of 0.863 in comparative experiments. These results confirm the contribution of a combined use of multi-scale features and attention to edge guided polyp boundary estimation when the boundary is challenging.
On the other hand, using a relatively backbone network like Res2Net-50 means a higher requirement of computations, comparing to these light-weight CNN model. Therefore, achieving similar boundary awareness with a smaller decoder remains an important research direction.

2.3.2.9 Study 9: FocusU^{2}Net with Dual Attention and Gated U-Net

FocusU^{2}Net introduced a bi-level nested U-shaped architecture combined with dual attention mechanisms for colonoscopic polyp segmentation [9]. The architecture aims at enhancement of feature extraction and selective feature fusion using nested U-Net structure and attention mechanism.
The model was tested on several polyp datasets with reports of improvement over several segmentation methods. The article states that the dice-score improvements ranged from 3.14% to 43.59% over the compared methods and that it showed good generalization with cross-validation. These results indicate that attention and nested encoder–decoder structures can improve segmentation quality, particularly for difficult polyp regions. However nested structures and multi attention heads add to the complexity and may not be ideal for devices with very tight resources.

2.3.2.10 Study 10: AdaptUNet with Wavelet Transformation and Attention

AdaptUNet was presented as a hybrid U-Net model employing wavelet transform along with attention mechanisms and skip connections for segmentation of colorectal polyps [10]. Wavelet transformation enables capturing information in frequency domain which may not be revealed via ordinary spatial convolution whileattention mechanism strengthens features by better merging information from shallow and deeper layers.
The model was validated using training data obtained from HyperKvasir and tests on CVC-300, Kvasir-SEG and ETIS-LaribDB datasets. On CVC-300, AdaptUNet achieved a Dice coefficient of 0.9104, IoU of 0.8368, and balanced accuracy of 0.9880. These results indicate that combining frequency-domain information with attention can improve segmentation quality [10]. Nevertheless, the inclusion of wavelet processing introduces additional architectural operations and may increase implementation complexity compared with a conventional CNN encoder and lightweight decoder.

2.3.2.11 Study 11: IRv2-Net with a Pretrained InceptionResNetV2 Encoder

Ahamed et al. proposed IRv2-Net, a U-Net-style segmentation architecture incorporating a pretrained InceptionResNetV2 encoder and test-time augmentation [11]. The pretrained encoder provides rich hierarchical representations, while the U-Net-style decoder reconstructs the high-resolution segmentation mask. This study is particularly relevant to the use of pretrained CNN encoders in polyp segmentation.
The model was evaluated on colorectal polyp datasets and demonstrated that a powerful pre-trained encoder can improve feature representation when the available polyp training data are limited. Test-time augmentation further improves prediction robustness. However, Inception-ResNetV2 is substantially heavier than ResNet-34, making it less suitable for lightweight or real-time deployment. Therefore, replacing the heavy encoder with a smaller pretrained residual backbone while retaining strong contextual and boundary modeling is a useful research direction.

2.3.2.12 Study 12: Meta-Polyp-An Efficient Polyp Segmentation Baseline

Trinh et al. introduced Meta-Polyp as an efficiency-oriented baseline for colorectal polyp segmentation [12]. The study investigated compact segmentation designs intended to provide a favorable balance between segmentation accuracy and computational requirements. The proposed model was tested on standard colorectal polyp databases, where it showed comparable results to large segmentation models.
The primary contribution of Meta-Polyp focuses on accurate vs efficient design trade-offs, but one caveat with lighter architectures can be an increased loss of some high level semantic representation when data is scarce. We address this by using a pre-trained feature extractor, such asResNet-34, to recover good semantic features without overly large segmenting networks.

2.3.2.13 Study 13: Polyp-PVT-Pyramid Vision Transformer for Polyp Segmentation

Dong et al. proposed Polyp-PVT, a pyramid vision transformer, for polyp segmentation [13]. The Polyp-PVT utilized hierarchical transformers to encode contextual information and employed similarity aggregation and camouflage attention to capture global contextual dependencies. This architecture may help the model to maintain effective relation at long distance and the capability of discriminate polyp with comparable tissue become improve. Polyp-PVT obtained good performance on several polyp segmentation benchmarks. Although transformers generally

achieve CNN comparable results only when a lot computational sources are invested. Hence Polyp-PVT is quite unsuitable for actual use given trade-off between speed and precision. The study therefore proposed to use an ASPP bottleneck to encode multi-scale context instead of using a large vision transformer encoder.

2.3.2.14 Study 14: PraNet-Parallel Reverse Attention Network

Fan et al. proposed PraNet, a parallel reverse attention network for polyp segmentation [14]. The attention mechanism allows for progressively refining segmentation predictions, emphasizing the most difficult regions to determine which were overlooked by coarse predictions. The network was designed to improve the boundary and region-level segmentation simultaneously.
PraNet achieved a mean Dice score of 0.898 and an mIoU of 0.840 on Kvasir-SEG in comparative benchmark evaluations. It also demonstrated strong performance on CVC-ClinicDB, with a reported mDice of approximately 0.899 and mIoU of 0.849. These results established reverse attention as an effective strategy for polyp segmentation [14]. However, the architecture is considerably more specialized than a lightweight U-Net and requires multiple refinement operations, which may increase computational complexity.

2.3.2.15 Study 15: ResUNet++ for Colorectal Polyp Segmentation

Jha et al. proposed ResUNet++, an improved residual U-Net architecture for medical image segmentation and specifically evaluated it for colonoscopic polyp segmentation [15]. The architecture combines residual blocks, squeeze-and-excitation mechanisms, ASPP, and attention-based feature processing within a U-Net-style framework. The combination of residual learning and multi-scale contextual information is directly relevant to architectures that use a residual encoder and ASPP bottleneck.
On Kvasir-SEG, ResUNet++ achieved a Dice coefficient of 81.33% and an mIoU of 79.27%. On the CVC-612 dataset, it achieved a Dice coefficient of 79.55% and mIoU of 79.62% [15]. Although the model improves upon conventional U-Net and ResUNet, its architecture contains several additional modules and therefore has a relatively high computational burden compared with lightweight networks. This motivates a more compact combination of pretrained residual encoding, ASPP-based context extraction, and efficient decoding.

2.3.2.16 Study 16: Attention U-Net and Attention Gates

Oktay et al. introduced Attention U-Net, which incorporated attention gates into the U-Net architecture to selectively emphasize relevant regions and suppress irrelevant background features [16]. The attention gates operate on encoder features before they are transferred through skip connections, allowing the decoder to receive more task-relevant information.
The architecture was tested on abdominal CT segmentation sets. On the TCIA pancreas set, the fine-tuned Attention UNet achieved a Dice score of 0.831 compared to 0.820 for the corresponding UNet, demonstrating the benefits of attention-based features selection. The attention gates also had relatively low computational overhead [16]. While this was not designed for colorectal polyps originally, attention-gating should be applied directly to polyps, e.g. Ignoring other mucosa before apply the skip-feature fusion.

2.3.2.17 Study 17: MobileNetV2 for Efficient Feature Extraction

Sandler et al. introduced MobileNetV2, an efficient CNN architecture based on inverted residual blocks and linear bottlenecks [17]. The design makes the computational cost low because it uses depth-wise convolution. But it keeps a strong feature representation. MobileNetV2 eventually became one of the critical backbones for lightweight medical segmentation architectures. The standard MobileNetV2 1.0 model has around 3.5M parameters and achieves roughly 72% of ImageNet top-1 accuracy depending on implementation and evaluation protocol. It is compact enough to be used in resource-constrained applications. However, MobileNetV2 was designed for image classification, and therefore additional decoder parts and multi-scale features mechanisms are required when adapting it to the dense pixel-level tasks like polyp segmentation. This limitation inspired the use of a lightweight convolutional principles within a U-Net decoder as opposed to using a full classification backbone.

2.3.2.18 Study 18: CBAM Convolutional Block Attention Module

Woo et al. introduced the Convolutional Block Attention Module (CBAM) which is a lightweight attention module that sequentially applies channel attention and spatial attention to convolutional feature maps [18]. CBAM was designed to improve feature representation with relatively small computational overhead and has subsequently been widely adopted in medical image segmentation.
Experiments on image classification and detection benchmarks demonstrated that CBAM improves representation quality by emphasizing informative channels and spatial regions. Its lightweight design makes it attractive for segmentation networks. However, CBAM primarily performs local channel and spatial refinement and does not directly provide the multi-scale contextual modeling obtained from an ASPP module or the selective skip-feature gating provided by Attention U-Net. Therefore, different attention mechanisms address complementary aspects of segmentation.

2.3.2.19 Study 19: MobileNets for Lightweight Convolution

Howard et al. introduced MobileNets, a family of lightweight CNN architectures based on depthwise-separable convolution [19]. Instead of performing a computationally expensive standard convolution, MobileNet separates spatial filtering and channel mixing into depthwise and pointwise operations. This substantially reduces the number of parameters and computational operations.
The MobileNet 1.0 model contains approximately 4.24M parameters and achieved 70.7% ImageNet top-1 accuracy at 224×224 resolution. The architecture was explicitly designed for low-latency and low-power applications, making its depthwise-separable convolution particularly relevant to lightweight segmentation decoders. However, MobileNet itself is primarily a classification architecture, and its use for segmentation requires an appropriate encoder–decoder design and high-resolution feature reconstruction.

2.3.2.20 Study 20: Xception and Deep Depthwise-Separable Convolution

Chollet introduced Xception, which further developed the idea of depthwise-separable convolution by decoupling spatial and cross-channel feature learning [20]. The architecture demonstrated that depthwise-separable convolution can provide strong representational performance while reducing computational requirements compared with conventional convolutional architectures.

Xception achieved competitive ImageNet classification performance while providing a more efficient convolutional design. The main limitation is that Xception remains a relatively deep architecture and is not itself designed specifically for medical image segmentation. Nevertheless, the depthwise-separable convolution principle allows one to replace computationally expensive standard convolutions with more efficient operations in segmentation decoders.

2.3.3 Comparative Analysis of Existing Studies

The reviewed studies have identified several trends in colorectal polyp segmentation. Firstly, the U-Net like encoder-decoder architecture still holds because the skip connection maintains the spatial information during the reconstruction. Secondly, the implementation of residual and pretrained encoders enhances the semantic representation, especially when the amount of polyp training data is limited. Third, attention mechanisms, including attention gates, channel/spatial attention, and edge-guided attention, improve the ability to focus on relevant polyp regions and ambiguous boundaries. Fourth, multi-scale contextual modules such as ASPP help represent polyps with substantial variation in size and appearance.
Lightweight architectures such as MobileNet, MobileNetV2, DeepNeXt, and Li-DeepLabV3+ demonstrate that computational efficiency can be improved substantially using depthwise-separable convolution and compact feature extraction.
At the same time, high-performing models such as Polyp-PVT and foundation-model-based approaches demonstrate that richer global representations can improve segmentation accuracy but usually introduce greater computational or training complexity. The recent BAR-UNet study is particularly relevant because it combines a pretrained ResNet-34 encoder with a U-Net-style decoder and explicitly addresses boundary quality, achieving a Dice score of 0.8798 on Kvasir-SEG [1]. Nevertheless, its boundary and appearance-consistency modules introduce additional complexity.

2.4 SUMMARY

Attention mechanisms enable the segmentation model to learn how to selectively focus on informative features while ignoring the irrelevant information during the encoding step, whilst ASPP facilitates multi-scale contextual representation. In addition, depthwise-separable convolution provides an efficient mechanism for feature reconstruction.
From above analysis of every technique, every method is trying to resolve some different kinds of problems for medical image segmentation. But can achieve the weaknesses through integrating the methods. Hence, this research proposes PRISM-UNet (Pyramid Residual Integration and Separable Multi-scale U-Net for colon polyp segmentation) model which is based on a pretrained ResNet34 encoder, ASPP-based multi-scale feature extraction, attention-gated skip connections, and depthwise-separable decoder.
In subsequent sections, the proposed model will be evaluated experimentally to determine its ability to perform accurate segmentation and to elucidate the contribution of each of the architectural components.

Chapter 3
Methodology

3.1 INTRODUCTION

Colorectal polyps are the abnormal cluster of cells that develop on the inner lining of the colon and may represent important predecessor to colorectal cancer. Therefore ,precise identification and delineation of polyps in colonoscopy images are essential for computer-aided diagnosis. However, due to the nature of polyp to show substantial variations in size, shape, texture, color, illumination, and boundary characteristics automatic polyp segmentation remains challenging. Additionally, colonoscopy images often contain complex background details, , low-contrast regions, specular reflections and irregular polyp boundaries, which can adversely affect segmentation performance accuracy.
This research explores an improved deep learning-based semantic segmentation framework for automatic colon polyp segmentation. The study is initiated with a conventional U-Net baseline and gradually incorporates architectural alteration to improve feature representation, contextual comprehension, feature selection, and computational efficiency. At first, A ResNet34 encoder initialized with ImageNet-pretrained weights is adapted for enhanced feature extraction. Then an Atrous Spatial Pyramid Pooling (ASPP) module is integrated at the bottleneck to capture multi-scale contextual information, after that to emphasize relevant polyp features and suppress background information attention gates are introduced in the skip connections. Furthermore, a depthwise-separable decoder is incorporated to reduce the computational complexity while maintaining effective feature reconstruction.
The consequent architecture merges four principal components: a pretrained ResNet34 encoder, attention-gated skip connections, ASPP-based bottleneck, and a depthwise-separable decoder. The models are then trained and evaluated using the Kvasir-SEG dataset which consists of 1,000 image-mask pairs. The dataset is divided into 3 parts- 800 training, 100 validation, and 100 test samples using a fixed random seed. All images are resized down to 256×256 pixels, with training-time augmentation and during preprocessing ImageNet normalization was applied.
Additionally, along with the primary model evaluation, the study integrates component-wise ablation experiments, qualitative failure-mode analysis, cross-dataset evaluation, and multi-seed robustness analysis. Also, an alternative MBLNet architecture is also probed for comparative evaluation. These experiments provide a methodical assessment of segmentation performance, architectural contribution, generalization ability, and robustness of the investigated models.

3.2 METHODOLOGY OVERVIEW

The overall experimental methodology follows a methodological and reproducible deep-learning pipeline consisting data acquisition and preprocessing, model development, training, in-distribution evaluation, hyperparameter optimization, cross-dataset evaluation, qualitative analysis, component-wise ablation, and multi-seed robustness analysis.
For the general experiments, random seed 42 is fixed to obtain a consistent 80% training, 10% validation, and 10% test split across the examined models. The same experimental constraints are maintained during model comparison to ensure a fair evaluation. Hyperparameter optimization is performed before final model training, and then the selected configuration is subsequently was used for the primary evaluation. The trained models are at first evaluated on the held-out

Kvasir-SEG test set using standard segmentation metrics. Their generalization capability is then assessed on independent external datasets without additional training. Component-wise ablation experiments are conducted to determine the contribution of the major architectural components, while qualitative analysis is used to examine representative successful and failure cases.

Finally, the robustness of the complete proposed framework is evaluated through a multi-seed analysis using six independent random seeds. The complete experimental workflow is illustrated in Figure **??**.

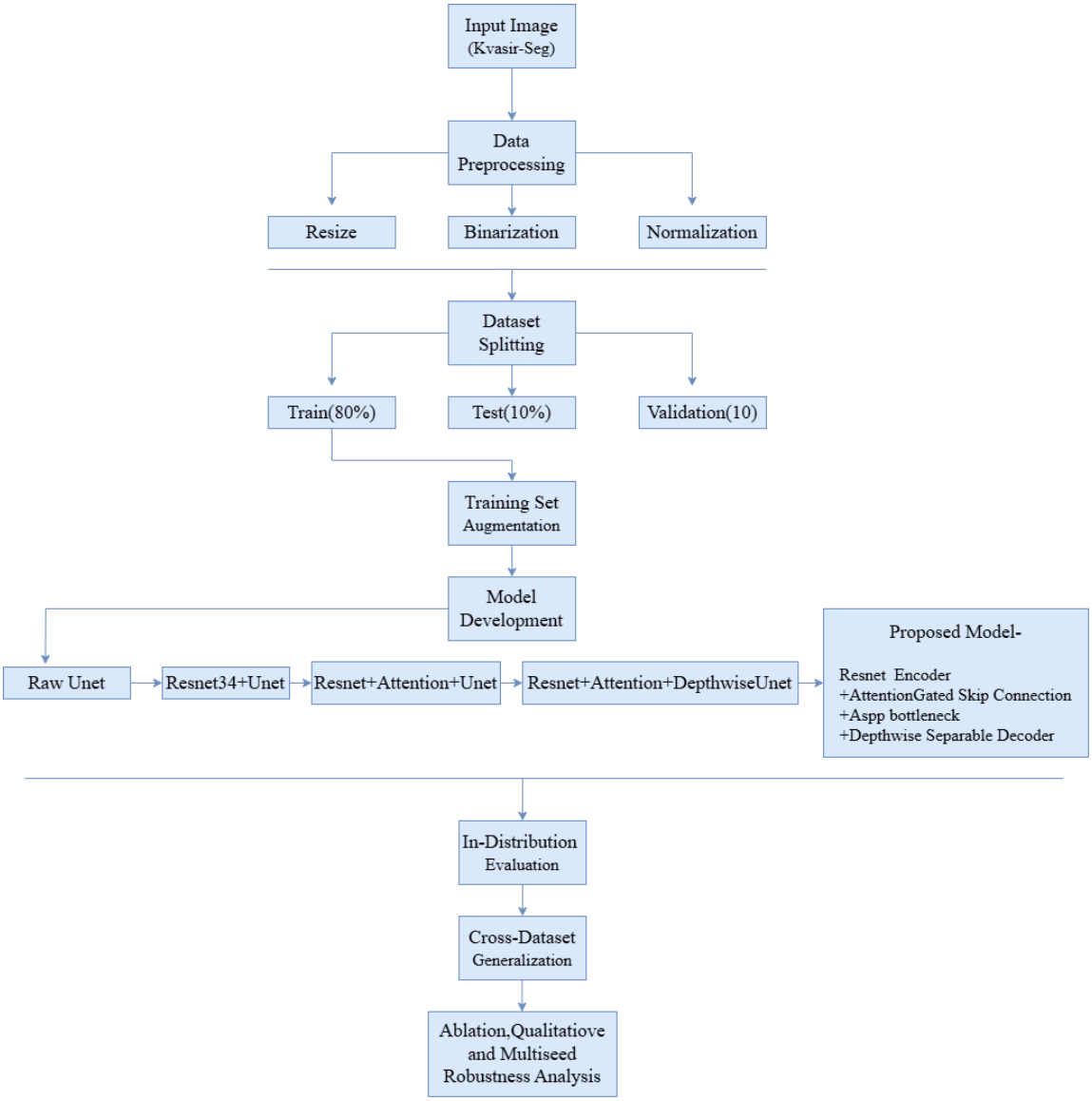


Figure 3.1: Flowchart of Overall Architecture

3.3 DATASET

3.3.1 Kvasir-SEG Dataset

The primary dataset used for training and in-distribution evaluation is Kvasir-SEG, a publicly available, expert-annotated dataset of gastrointestinal polyp images and their corresponding binary segmentation masks. Each image has a pixel-aligned ground-truth mask indicating the polyp region. Images were read as RGB and paired with their corresponding masks by filename. The dataset contains 1,000 image-mask pairs. Because the dataset does not provide patient or case identifiers, the experiments use file-level random splitting. The primary ResNet34 experiments use a fixed-seed 80%/10%/10% training-validation-test split. MBLNet uses an equivalent split generated through a two-stage stratified random split.
Three independent datasets are additionally used exclusively for zero-shot cross-dataset evaluation:

- CVC-ClinicDB;
- CVC-ColonDB;
- ETIS-LaribPolypDB.

These datasets are not used for retraining or fine-tuning the proposed model.

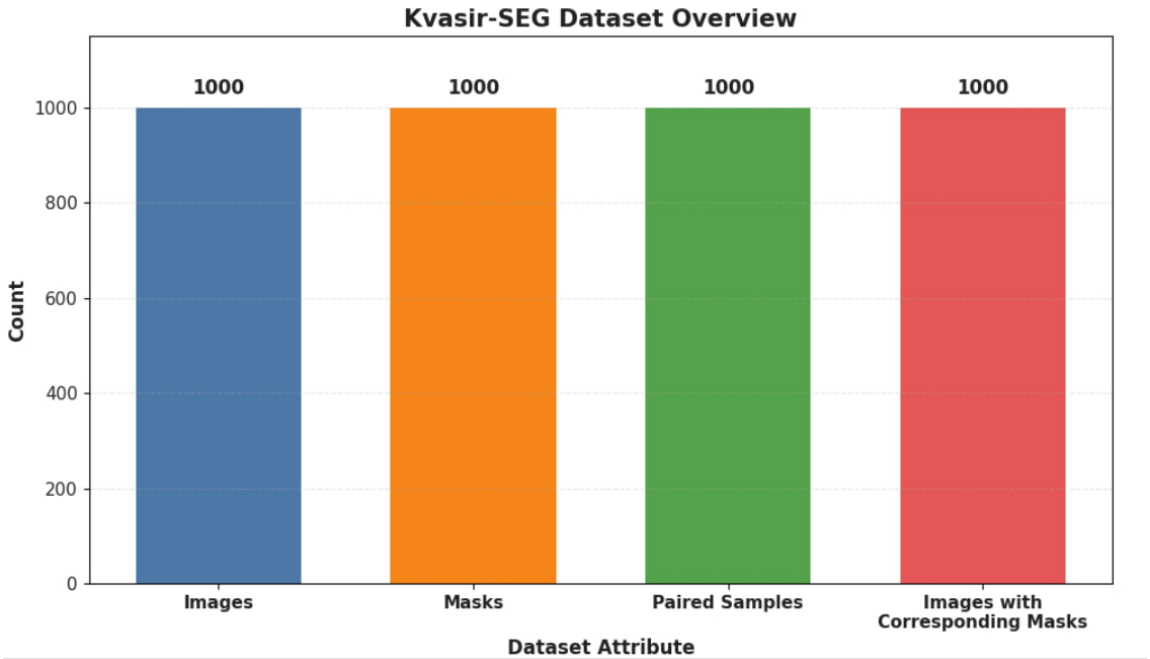


Figure 3.2: Dataset Description Barchart

3.3.2 Sample Dataset

1) Data Overview:

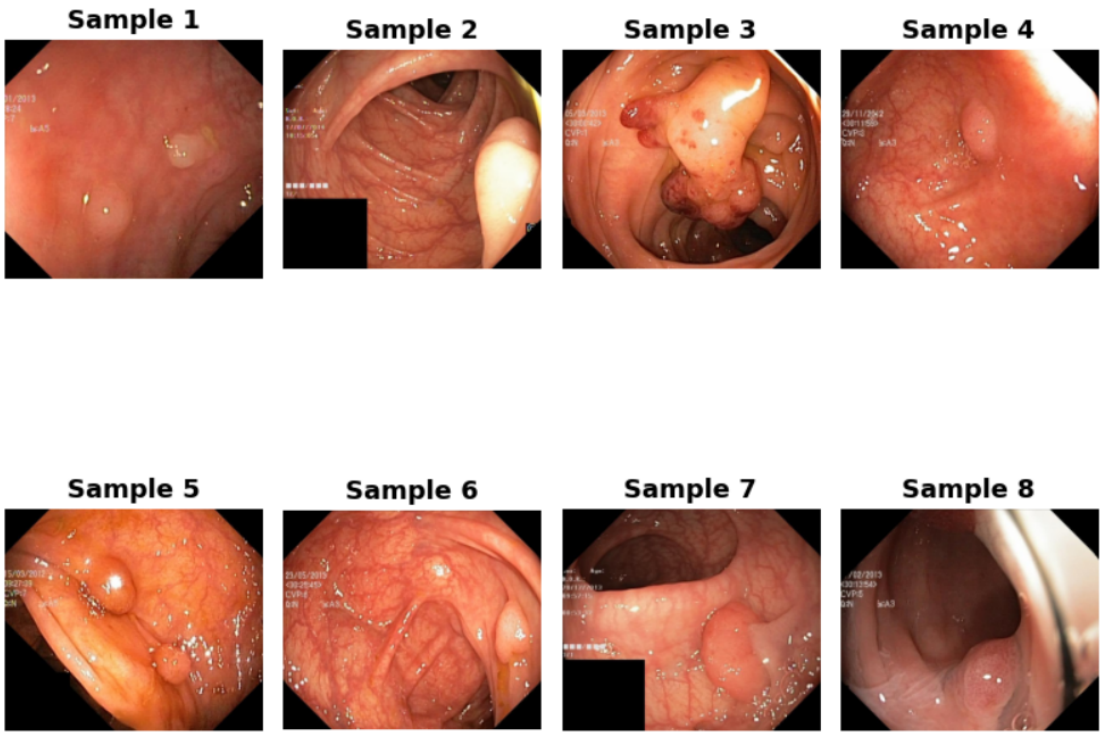


Figure 3.3: Sample Polyp images

2)Data with mask :

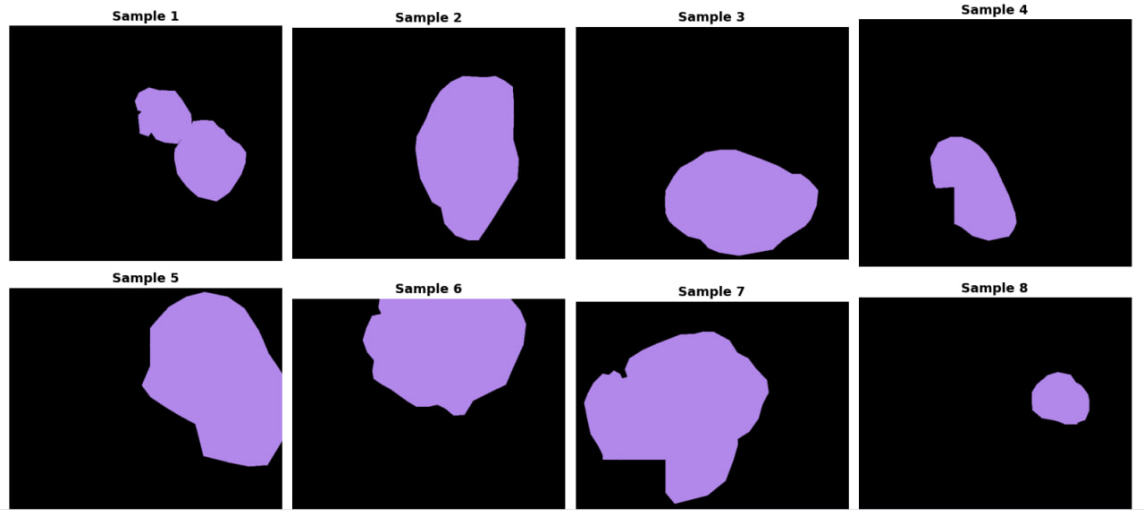


Figure 3.4: Sample mask image of polyps

3)Data with mask and overlay:

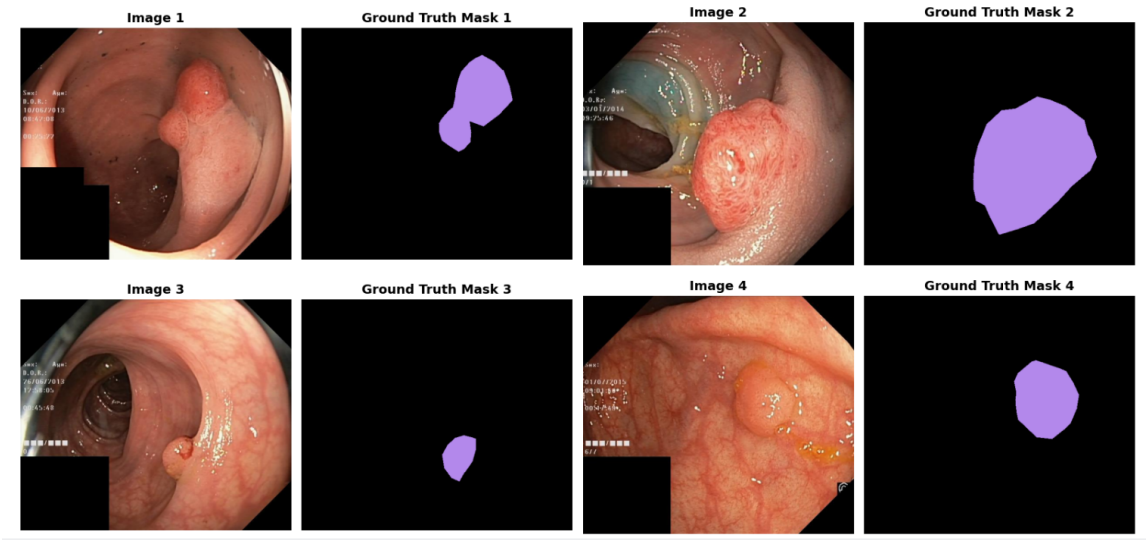


Figure 3.5: Sample polyp image with corresponding mask

3.3.3 Dataset Preparation

The Kvasir-SEG dataset was randomly divided using a fixed random seed. The resulting distribution consisted of 800 training images, 100 validation images, and 100 testing images. Data augmentation was applied only to the training set, while validation and testing sets remained unchanged.

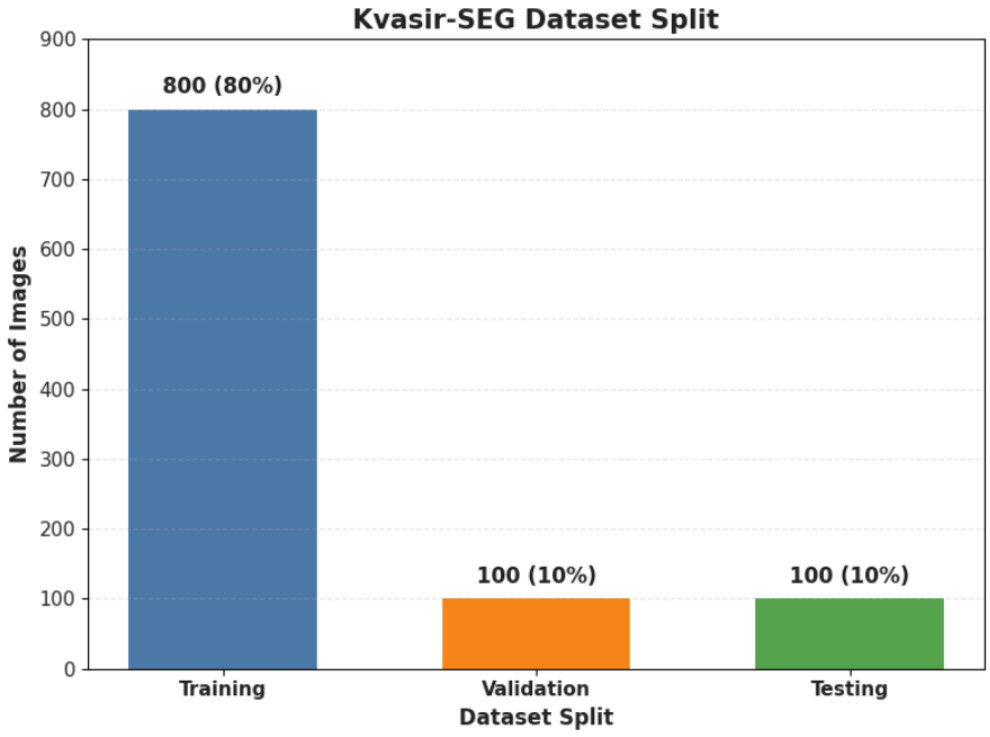


Figure 3.6: Distribution of Kvasir-SEG images across training, validation, and testing subsets.

3.3.4 Image Preprocessing

All images were resized to 256×256 pixels. Bilinear interpolation was used for RGB images, whereas nearest-neighbor interpolation was used for segmentation masks to preserve their binary labels.
The grayscale ground-truth masks were converted into binary masks using a threshold of 127. Pixels with intensity values greater than 127 were assigned to the polyp class (1), while the remaining pixels were assigned to the background class (0).
Since the encoder uses ImageNet-pretrained ResNet34 weights, the input images were normalized using the ImageNet mean and standard deviation. Finally, the normalized images were converted into tensors using channel-first $(C \times H \times W)$ ordering.

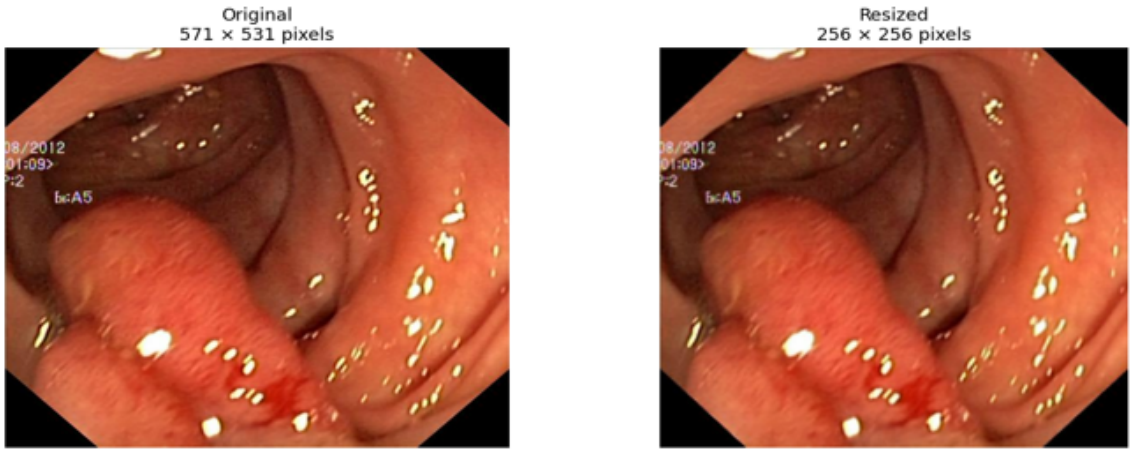


Figure 3.7: Sample 256*256 resized image

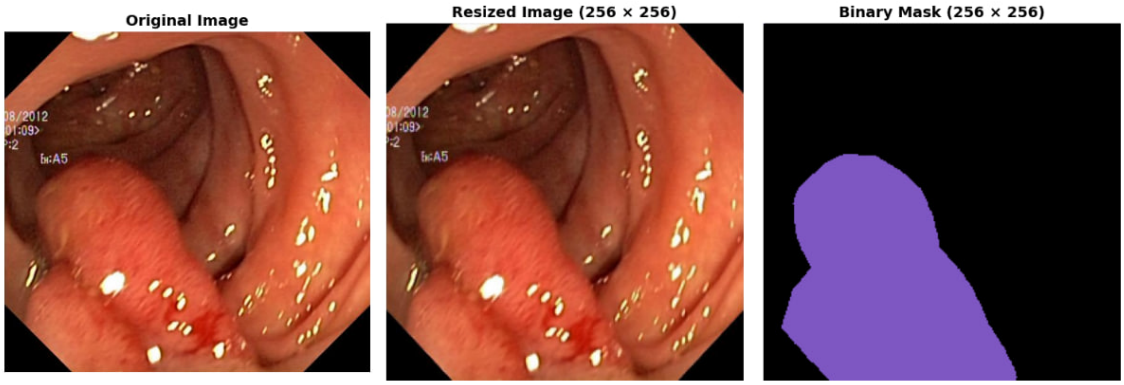


Figure 3.8: Sample 256*256 resized image with corresponding mask

3.3.5 Data Augmentation

Data augmentation was applied only to the training set to increase sample diversity and reduce overfitting. The augmentation pipeline included horizontal flipping, vertical flipping, random $90°$, $180°$, and $270°$ rotations, and random brightness/contrast adjustments, each with a probability of 0.5.
Geometric transformations were applied identically to images and masks to preserve pixel-level alignment. Brightness and contrast transformations were applied only to the input images.

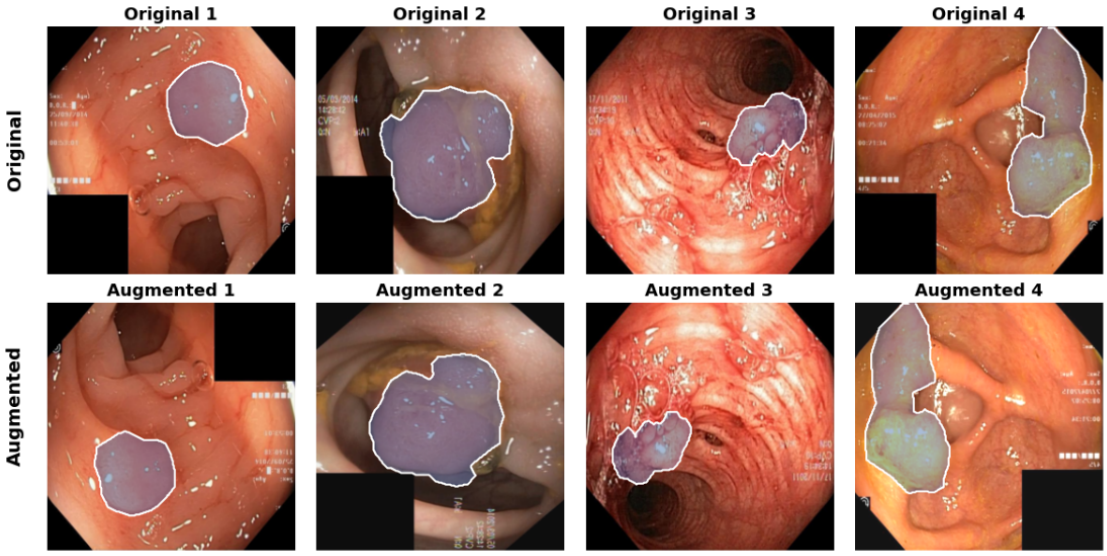


Figure 3.9: Sample resized image with mask

3.3.6 Training Data Augmentation

The original training dataset consisted of 700 samples. An additional 700 samples were generated through augmentation, resulting in a total of 1,400 training samples.
As illustrated in Figure 3.10, the number of training samples increased from 700 before augmentation to 1,400 after augmentation. This two-fold increase provides greater variability in the training data and helps the segmentation network learn more robust feature representations. Consequently, augmentation can reduce the risk of overfitting and improve the model's generalization capability when processing unseen images.

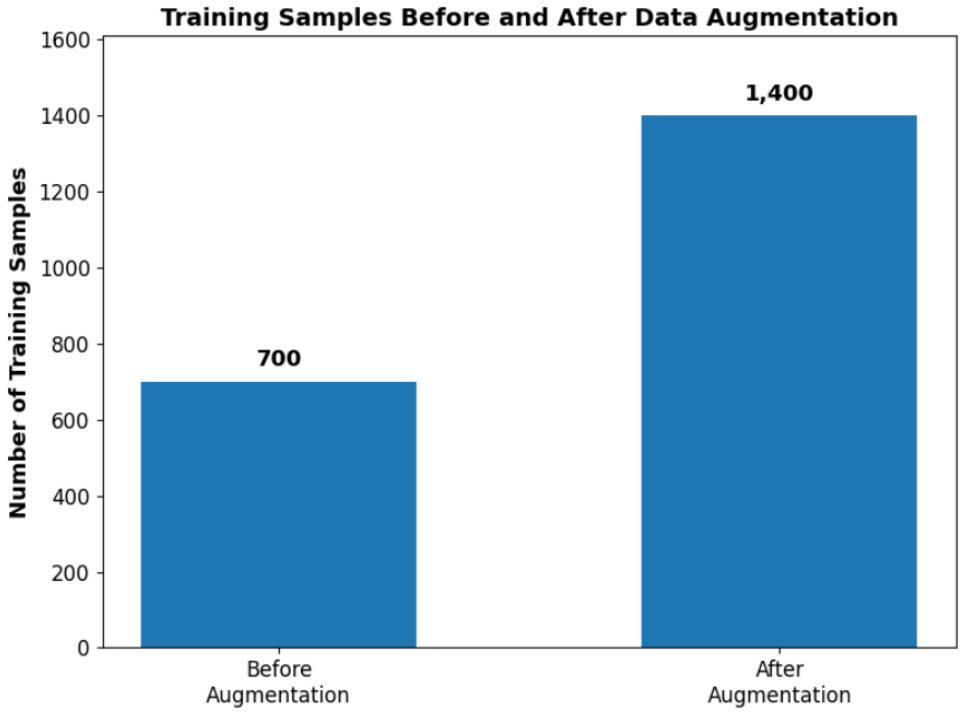


Figure 3.10: Training samples before and after data augmentation.

3.4 BASELINE MODEL DEVELOPMENT

At first, the conventional U-Net architecture was adopted as the baseline segmentation Model for this research . U-Net follows an encoder-decoder structure in which the encoder gradually extracts hierarchical depictions from the input image, while the decoder rebuilds the spatial details to generate the final segmentation map.
Then skip connections are used to transfer high-resolution features from the encoder to the corresponding decoder stages, allowing the network to retain spatial informations that may be lost during down-sampling of the input image. Therefore, the baseline U-Net provides a reference architecture against which the effectiveness of subsequent modifications can be evaluated.

3.4.1 Baseline U-Net

The standard U-Net consists of a contracting encoder path and an expanding decoder path connected through symmetric skip connections. The encoder progressively reduces spatial resolution while increasing the number of feature channels to extract increasingly abstract representations. The decoder subsequently restores spatial resolution through upsampling and convolutional operations.
The baseline architecture can be expressed as:

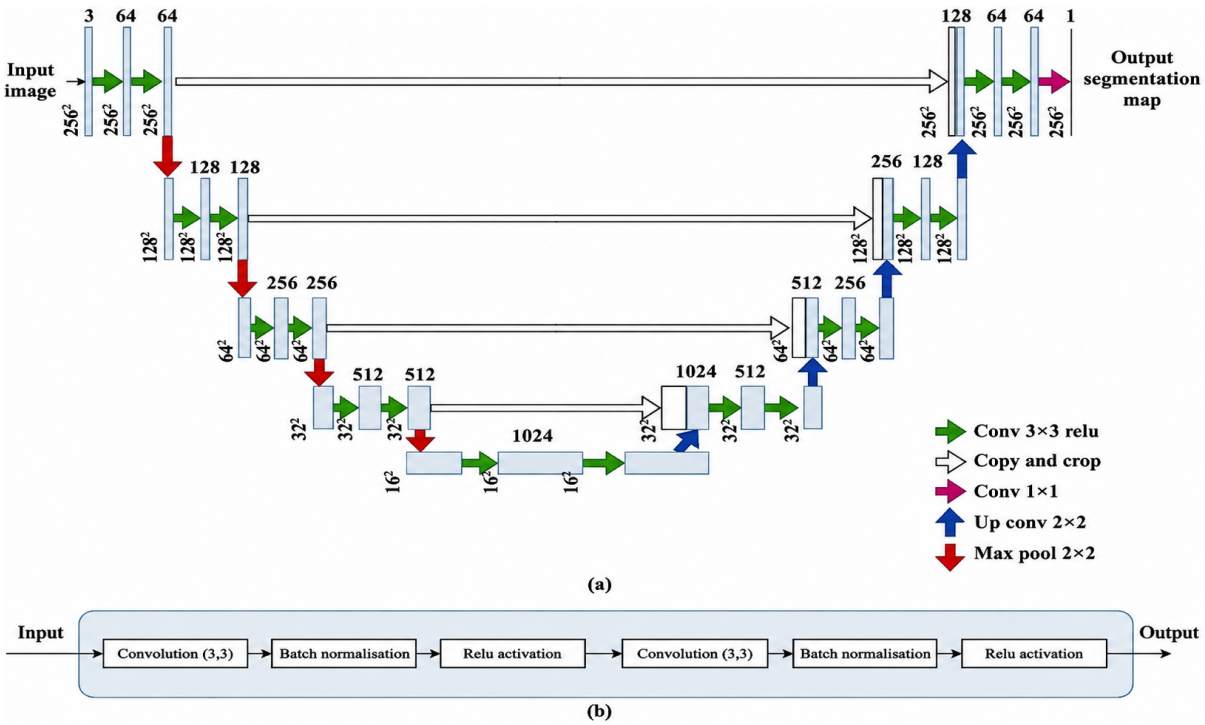


Figure 3.11: Architecture of the Baseline Unet [25]

3.4.2 ResNet34 Encoder

To improve the feature extraction capability of the baseline U-Net, its conventional encoder is replaced with a pretrained ResNet34 network. ResNet34 employs residual learning through identity shortcut connections, allowing deeper feature representations to be learned while reducing the degradation problem associated with conventional deep networks. The encoder

is initialized with ImageNet-pretrained weights, providing robust low and high level visual representations. In the implemented architecture, the ResNet34 encoder consists of the initial convolutional stem followed by four residual stages (Layer1-Layer4). The spatial resolution is progressively reduced from 256×256to 8×8, while the feature channels increase from 64 to 512. The resulting 8×8×512representation is passed to the bottleneck before decoding.

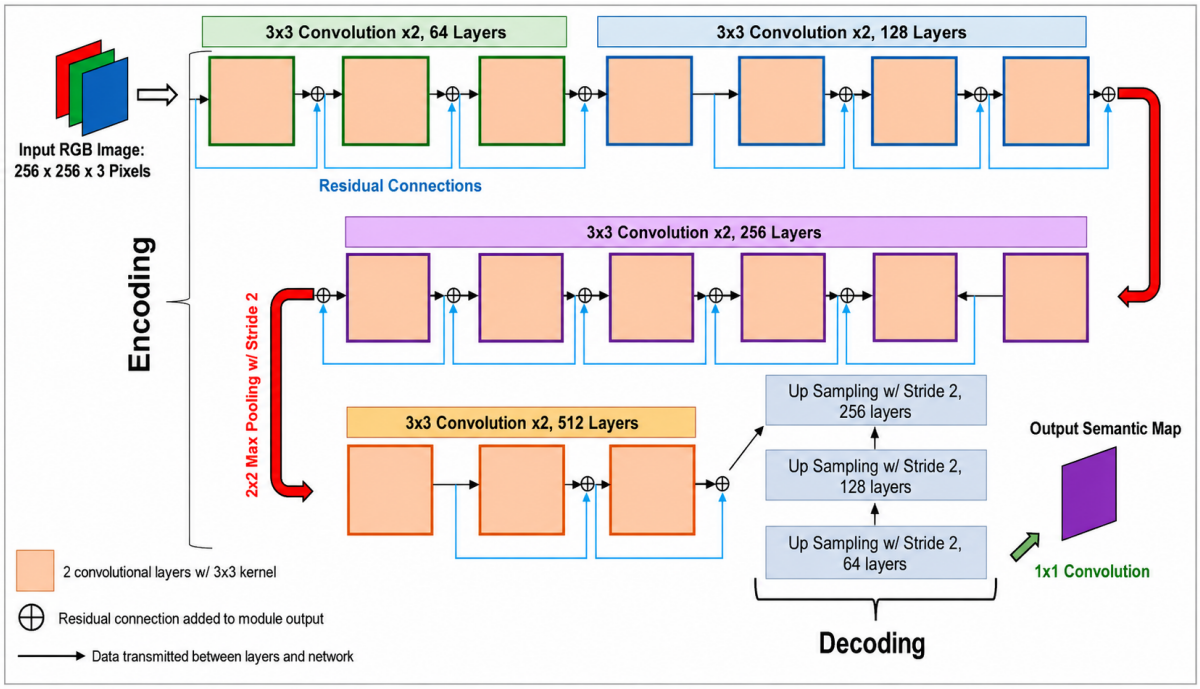


Figure 3.12: Architecture of the Baseline Resnet34 [26]

3.4.3 Resnet encoded Unet Architecture

The ResNet34-Encoder U-Net architecture replaces the conventional convolutional encoder of the standard U-Net with a pretrained ResNet34 network. ResNet34 is selected because its residual learning mechanism enables effective training of deeper networks while reducing the degradation problem associated with conventional deep convolutional architectures. The encoder progressively extracts hierarchical features from the input image, ranging from low-level details such as edges and textures to high-level semantic representations of the polyp regions.

The feature maps generated at different stages of the ResNet34 encoder are transferred to the corresponding decoder stages through skip connections. These connections preserve important spatial and boundary information that may be lost during downsampling. The decoder then progressively restores the spatial resolution of the feature maps through upsampling and convolutional operations. By combining deep semantic information from the encoder with fine-grained spatial information through skip connections, the architecture produces more accurate pixel-level segmentation maps.

The use of a pretrained ResNet34 encoder also provides stronger feature representations compared with the conventional U-Net encoder, while maintaining the overall encoder-decoder structure of U-Net. This architecture therefore serves as an improved baseline for subsequent integration of attention mechanisms, multi-scale feature extraction, and lightweight convolutional components in the proposed PRISM-UNet framework.

The ResNet34 encoder therefore replaces the original U-Net encoder while retaining the overall encoder–decoder segmentation framework.

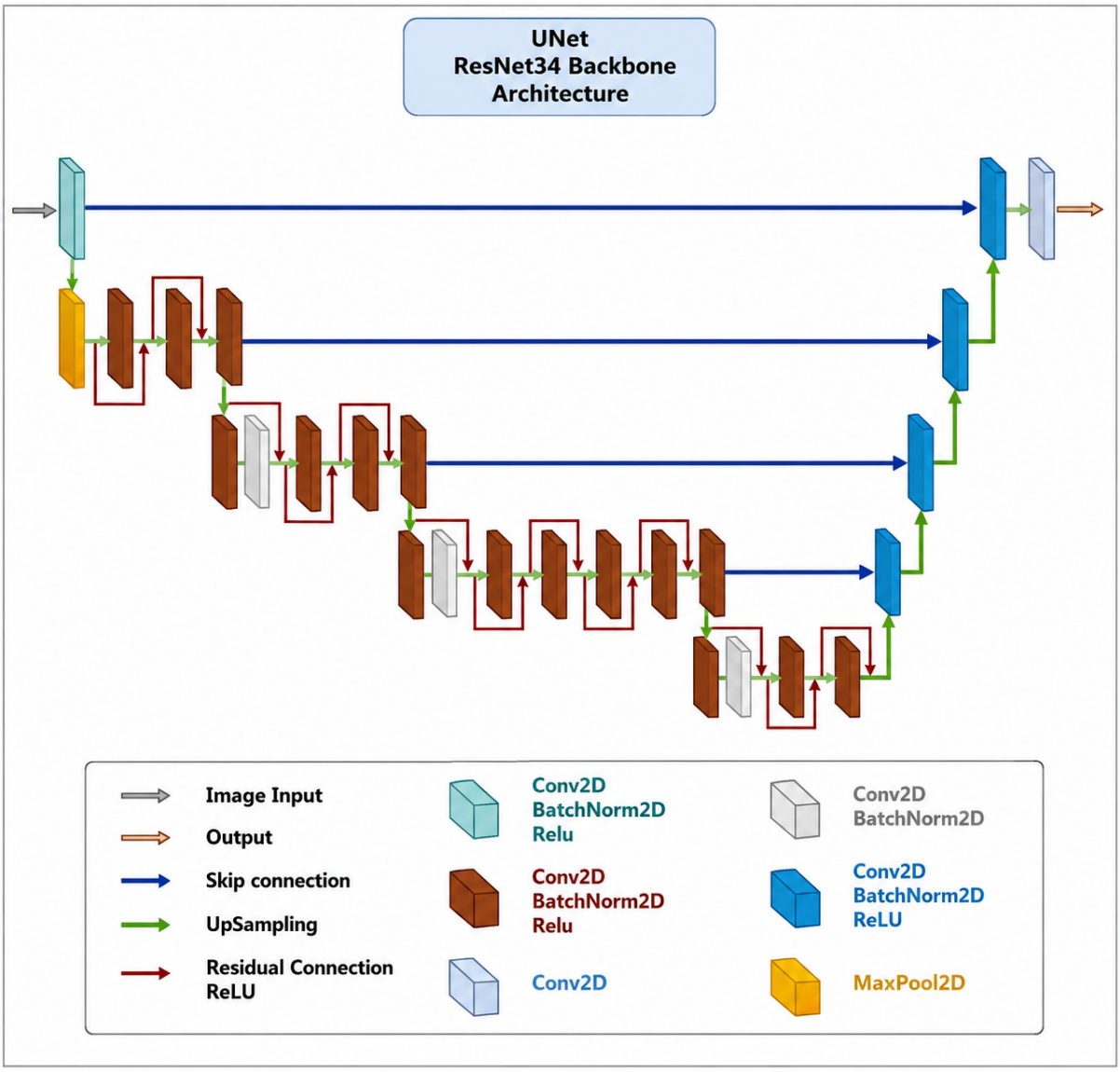


Figure 3.13: Architecture of ResNet34 encoded UNet [27]

3.4.4 Resnet encoded Unet Architecture with attention

Standard U-Net skip connections directly concatenate encoder features with decoder features. Although this preserves spatial information, the transferred features may also contain irrelevant or background information. To address this limitation, attention gates are incorporated into the skip connections. The attention gate receives two inputs: the encoder feature map and the gating signal generated by the decoder after upsampling. Both inputs are projected using 1×1convolutions and combined through an additive operation followed by ReLU activation. A subsequent 1×1convolution and sigmoid function generate an attention coefficient map. This coefficient map is used to selectively weight the encoder feature map before it is concatenated with the decoder features. The attention mechanism enables the network to emphasize informative regions while suppressing less relevant features in the skip connections. The attention resnet can be represented as:

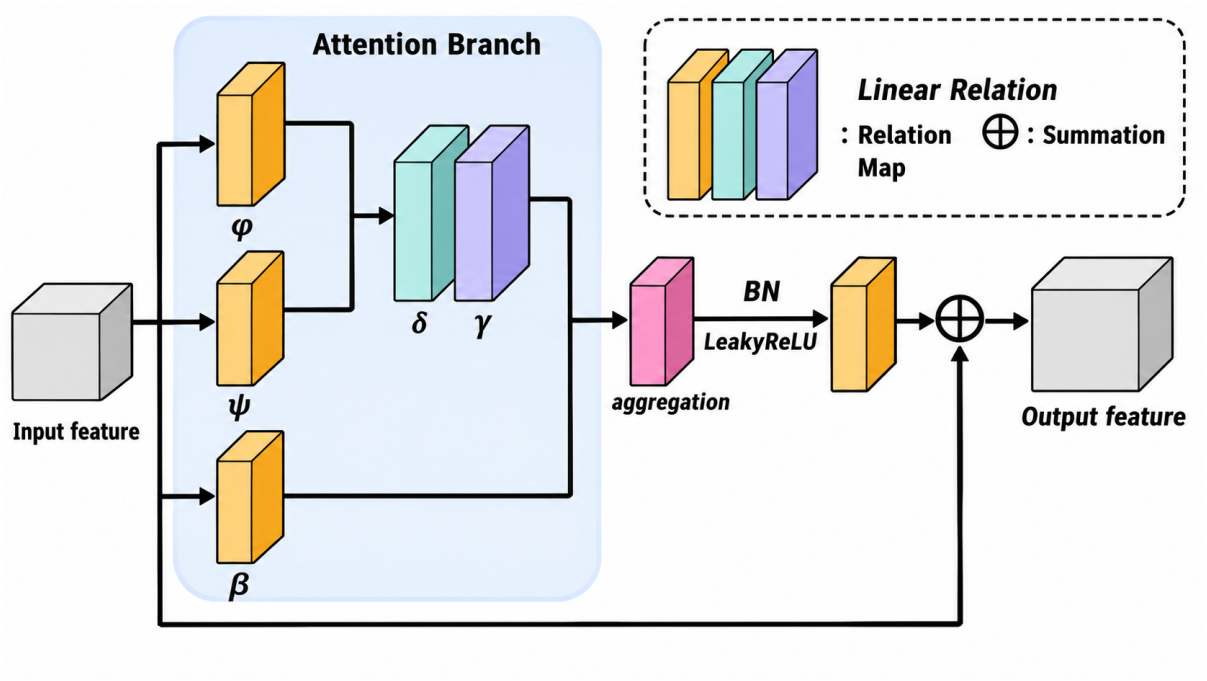


Figure 3.14: Sample Attention Module Architecture [28]

3.4.5 ResNet34 + Attention

The attention mechanism is subsequently incorporated into the ResNet34-based U-Net to evaluate its contribution independently. In this model, the pretrained ResNet34 encoder is retained, while attention gates are introduced into the skip connections. The decoder otherwise follows the conventional convolutional design.

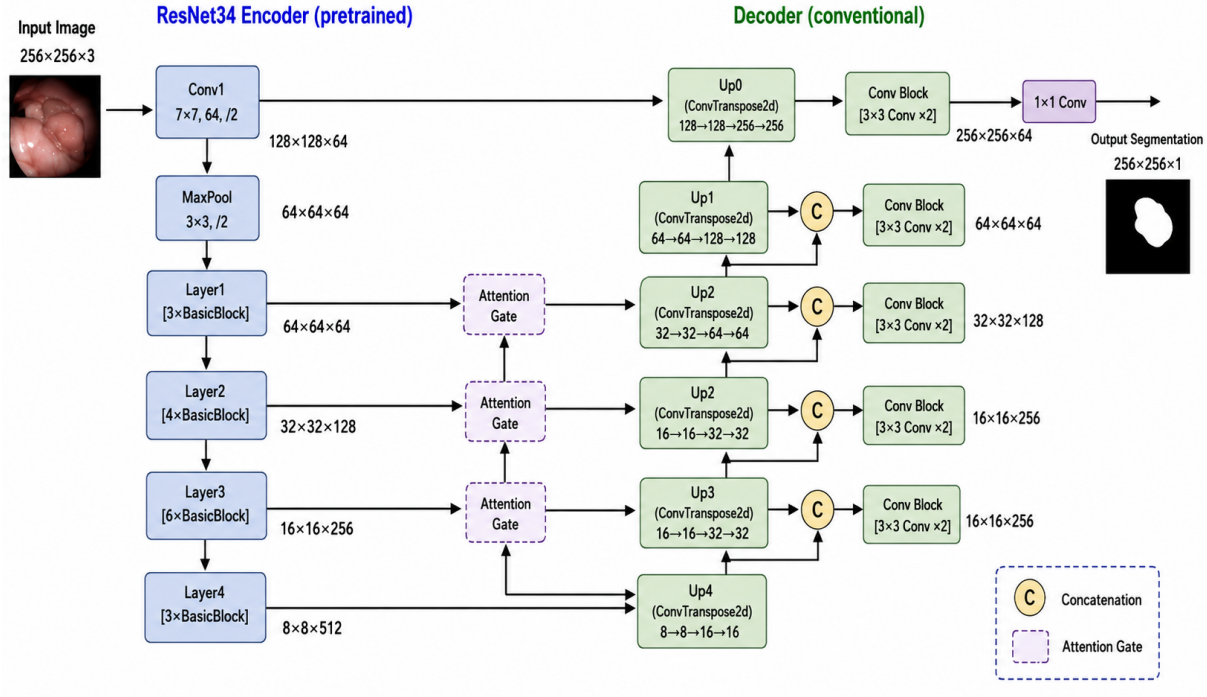


Figure 3.15: resnet Encoder with Attention Gated Skip Connection Unet

3.4.6 Depthwise-Separable Decoder

The conventional U-Net decoder relies on standard 3×3convolutions, which can introduce a considerable number of parameters and computational operations. To obtain a more computationally efficient decoder, standard convolutions are replaced with depthwise-separable convolutions. A depthwise-separable convolution decomposes a conventional convolution into two operations. First, a depthwise 3×3convolution independently processes each input channel. This is followed by a pointwise 1×1convolution that combines information across channels. In the implemented DSConv block, each operation is followed by batch normalization and ReLU activation. Two consecutive DSConv operations form the DSDoubleConv block used throughout the decoder. This design reduces the computational burden of the decoder while retaining the ability to perform spatial and channel-wise feature transformations.
The depthwise-separable convolution can be represented as:
The depthwise-separable decoder is then combined with the pretrained ResNet34 encoder to form an intermediate model. In this configuration, the ResNet34 encoder extracts hierarchical features, while the decoder uses transposed convolution for upsampling followed by DSDoubleConv blocks for feature refinement. Skip connections are retained between corresponding encoder and decoder levels.

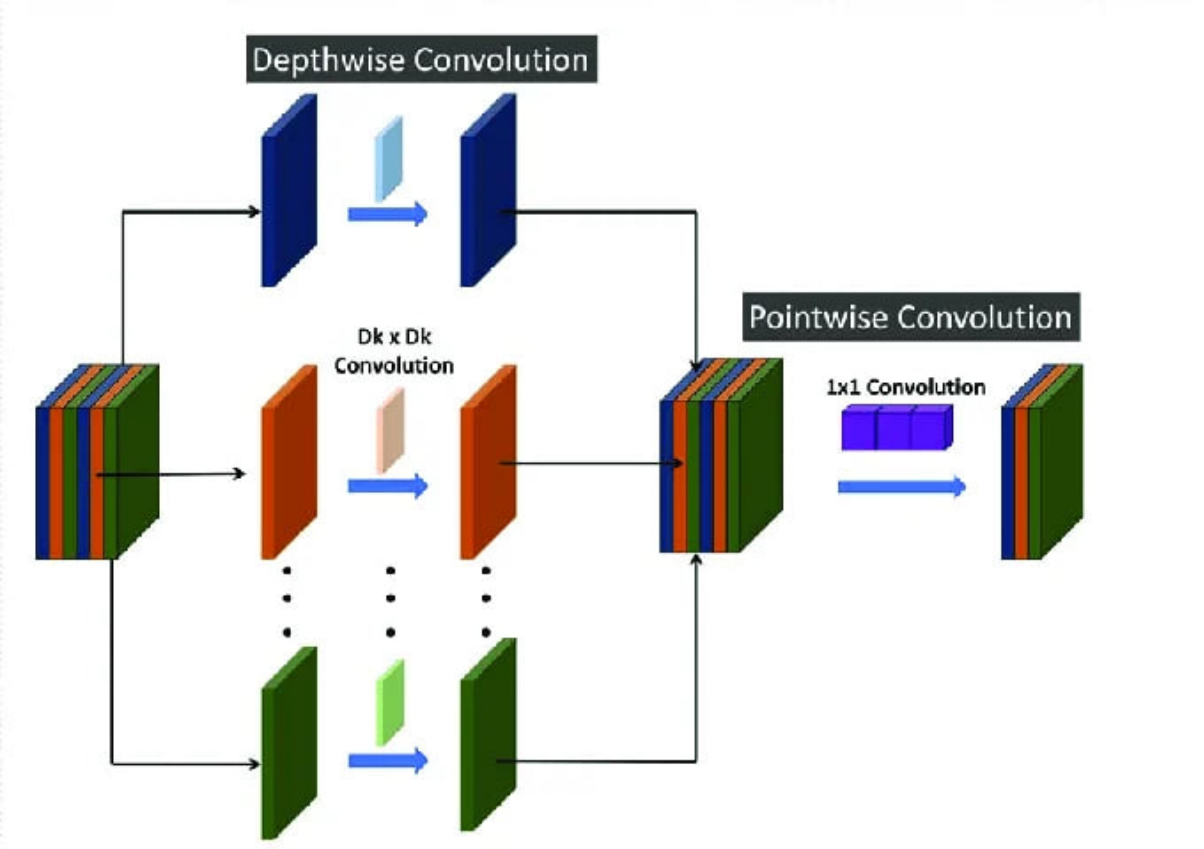


Figure 3.16: Depthwise-Separable-Convolution module [29]

The architecture combines a pretrained ResNet34 encoder with a lightweight depthwise-separable convolution decoder. ResNet34 extracts hierarchical features, while skip connections preserve spatial details. The depthwise-separable decoder reduces computational complexity and parameters while effectively reconstructing high-resolution segmentation maps.
The ResNet34 + Depthwise-Separable Decoder contains:

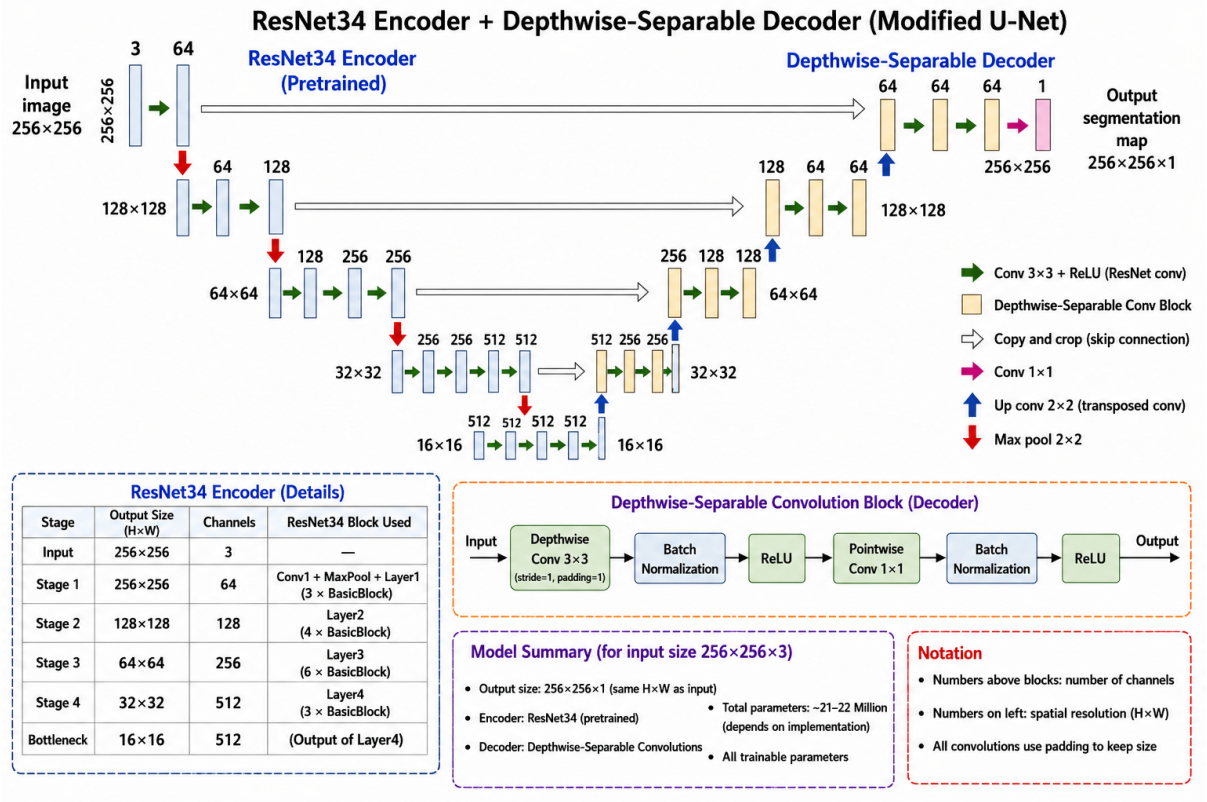


Figure 3.17: Resnet encoded depthwise separable decoder Unet

3.4.7 ASPP Multi-Scale Bottleneck

Although encoder features provide increasingly abstract representations at deeper levels, a single bottleneck representation may not sufficiently capture objects or structures at different spatial scales. Therefore, an Atrous Spatial Pyramid Pooling (ASPP) module is introduced at the bottleneck of the proposed network. The implemented ASPP receives the 8×8×512output of the ResNet34 encoder and extracts multi-scale contextual information through parallel branches. These branches consist of a 1×1convolution, three depthwise-separable atrous convolution branches with dilation rates of 6, 12, and 18, and a global average-pooling branch. The outputs of these branches are resized when necessary, concatenated, and projected through a 1×1convolution. Batch normalization, ReLU activation, and dropout are subsequently applied to the projected representation. The ASPP module enables the bottleneck to incorporate contextual information from multiple receptive-field sizes before the decoding process begins.

The ASPP bottleneck contains:

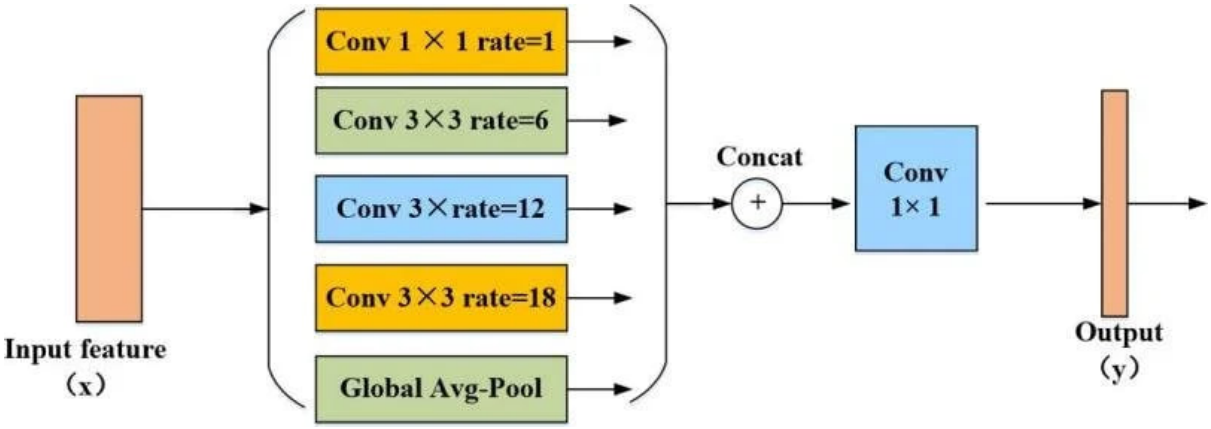


Figure 3.18: ASPP module architecture [30]

3.4.8 Proposed model

Based on the preceding modifications, the final proposed PRISM-UNet architecture integrates all four components within a unified U-Net framework to improve feature extraction, boundary localization, multi-scale contextual representation, and computational efficiency. The conventional U-Net encoder is replaced with a pretrained ResNet34 encoder, which utilizes residual learning to extract increasingly complex and discriminative features from the input image. The pretrained weights provide a strong initial representation and allow the encoder to capture both low-level structural details and high-level semantic information relevant to polyp segmentation. To improve the transfer of useful features from the encoder to the decoder, attention gates are incorporated into the skip connections. These gates selectively emphasize relevant polyp-related features while suppressing irrelevant background information. This selective feature fusion is particularly beneficial for colonoscopy images, where polyps may have low contrast, irregular shapes, or boundaries that are difficult to distinguish from surrounding tissues.

At the bottleneck, an Atrous Spatial Pyramid Pooling (ASPP) module is introduced to capture contextual information at multiple receptive-field scales. By applying parallel atrous convolutions with different dilation rates, ASPP enables the network to identify both local details and broader contextual patterns. The resulting multi-scale representation strengthens the model's ability to handle polyps with substantial variations in size and appearance.

The decoder employs transposed convolution for progressive spatial upsampling and depthwise-separable convolution blocks for feature refinement. Depthwise-separable convolutions decompose standard convolution into depthwise and pointwise operations, substantially reducing the number of parameters and computational operations while maintaining effective feature extraction. At each decoder stage, the upsampled features are combined with the corresponding attention-refined encoder features through skip connections, allowing the network to recover fine-grained spatial information and preserve accurate polyp boundaries.

The complete network accepts a $256 \times 256 \times 3$ RGB input image. The ResNet34 encoder progressively generates feature representations at spatial resolutions of 256×256, 128×128, 64×64, 32×32, 16×16, and 8×8, with the deepest representation containing 512 channels. The 8×8 feature representation is processed by the ASPP module to obtain multi-scale contextual features before being passed to the decoder. The decoder then progressively restores the spatial resolution through four upsampling stages, with attention-gated skip connections providing corresponding encoder features at each stage.

Finally, the refined decoder representation is passed through a 1×1 convolution that maps the 32-channel feature representation to a single output channel. Consequently, the network produces a $256 \times 256 \times 1$ segmentation logit map corresponding to the input image resolution. The final architecture therefore combines the strong residual feature extraction capability of ResNet34, selective feature fusion through attention gates, multi-scale contextual modeling through ASPP, and computationally efficient decoding through depthwise-separable convolutions. Together, these components are designed to provide accurate and efficient pixel-level segmentation of colorectal polyps, particularly for challenging cases involving variations in size, shape, appearance, and boundary definition.
The proposed architecture contains:

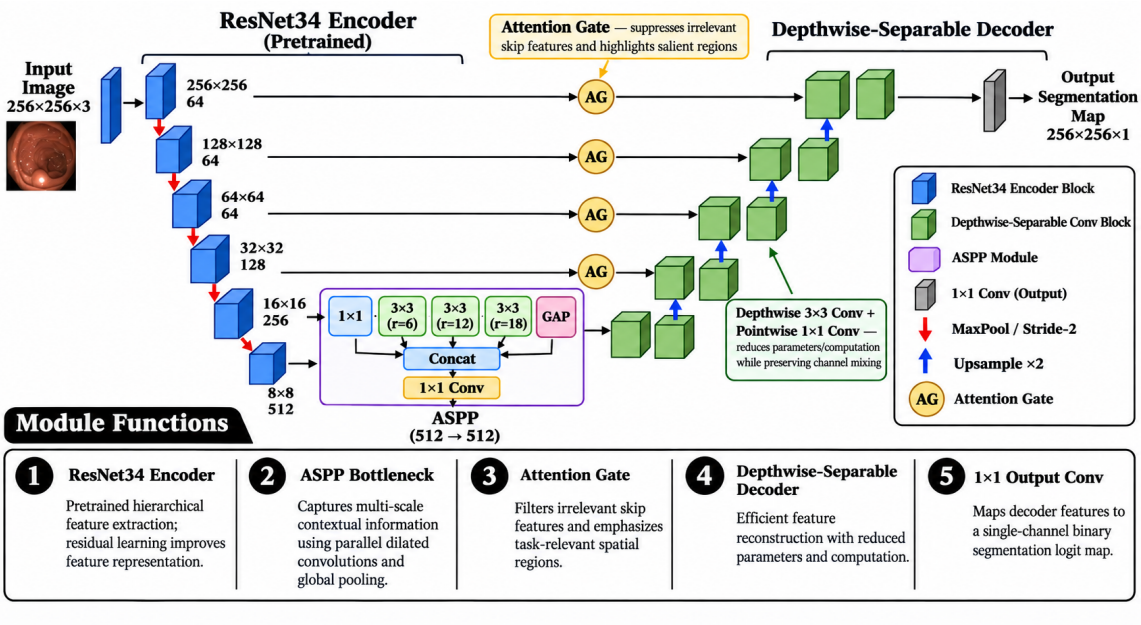


Figure 3.19: Proposed PRISM-Net model architecture

3.5 Architectural Progression

The architectural progression used to develop the final model is summarized in Figure 3.20. The model was developed incrementally from a conventional U-Net baseline by introducing the ResNet34 encoder, attention gates, ASPP-based multi-scale feature extraction, and a depthwise-separable decoder.

Table 3.1: Architectural progression of the proposed model.

Model	ResNet34 Encoder	Attention Gates	ASPP	Depthwise-Separable Decoder
Standard U-Net	No	No	NO	No
ResNet34-U-Net	Yes	No	No	No
ResNet34 + Attention	Yes	Yes	No	No
ResNet34 + Depthwise Decoder	Yes	No	No	Yes
Proposed Model	Yes	Yes	Yes	Yes

This progressive design allows the effect of the individual architectural modifications to be investigated through controlled model comparisons, while the final proposed model combines the complementary components into a single segmentation framework.

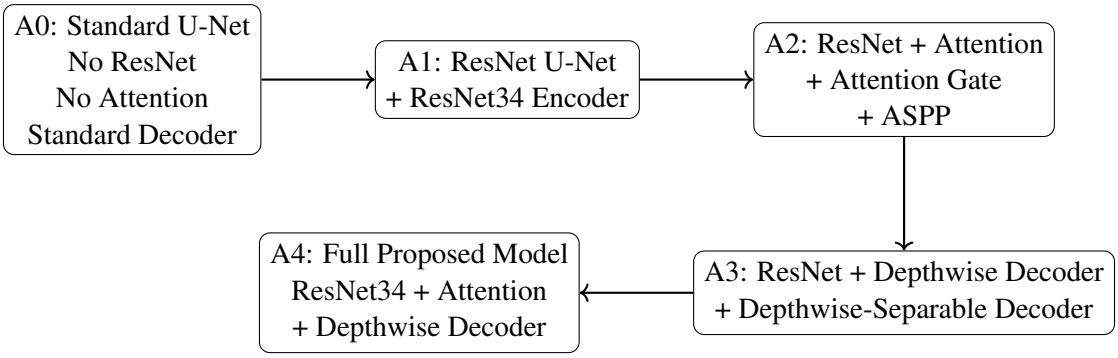


Figure 3.20: Progressive architectural design of the ResNet34-based U-Net architecture.

3.5.1 Loss Function

The segmentation network was trained using a combined Binary Cross-Entropy (BCE) and Dice loss to provide both pixel-level supervision and region-level overlap optimization. The combined objective is particularly suitable for medical image segmentation, where accurate identification of the target region and preservation of its spatial extent are both important.

3.5.1.1 Binary Cross-Entropy Loss

Binary Cross-Entropy loss measures the pixel-wise difference between the predicted probability and the corresponding ground-truth label. It is defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{3.1}$$

where N denotes the total number of pixels, y_i represents the ground-truth label of pixel i, and $\hat{y}_i$ represents the predicted probability of that pixel belonging to the foreground class.
BCE provides a stable pixel-level optimization signal and encourages the model to correctly classify individual pixels as foreground or background.

3.5.1.2 Dice Loss

To directly optimize the overlap between the predicted segmentation and the ground-truth mask, Dice loss is incorporated. It is defined as:

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i + \epsilon}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i + \epsilon} \tag{3.2}$$

where y_i and $\hat{y}_i$ is denoting the ground-truth label and predicted probability, individually, and ϵ is a small smoothing constant presented to enhance numerical stability.
Unlike BCE, which only evaluates pixels independently, Dice loss measures the exact overlap between the predicted foreground region and the ground-truth segmentation. This makes it specifically useful when the foreground occupies a relatively small or variable portion of the input image.

3.5.1.3 Combined Loss

The final training objective is obtained by combining the two loss components:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \mathcal{L}_{Dice} \tag{3.3}$$

The BCE component provides stable pixel-wise gradients, while the Dice component emphasizes region-level overlap. Their combination therefore provides a balance between accurate pixel classification and accurate foreground-region delineation, which is desirable for polyp segmentation where the target region can vary substantially in size, shape, and location within an endoscopic image.

3.5.2 Optimization and Training Strategy

The proposed model was optimized using the Adam optimizer with an initial learning rate of 2e-4 and weight decay of 5e-4. Training was performed with a batch size of 8 for a maximum of 40 epochs.
The validation Dice score was used as the primary criterion for model selection. A ReduceLROnPlateau scheduler reduced the learning rate by a factor of 0.5 when validation Dice stopped improving. Early stopping with a patience of 6 epochs was applied to prevent unnecessary training.

3.5.2.1 Training Configuration

Table 3.2: Training configuration of the proposed PRISM-Net model.

Parameter	**Setting**
Optimizer	Adam
Initial learning rate	2e-4
Weight decay	5e-4
Batch size	8
Maximum epochs	40
Learning-rate scheduler	ReduceLROnPlateau
LR reduction factor	0.5
Model-selection criterion	Validation Dice
Early-stopping patience	6 epochs
Actual training duration	29 epochs
Best validation Dice	0.9140

3.5.3 Performance Evaluation

The trained models were evaluated using multiple segmentation metrics to provide a comprehensive assessment of region overlap, pixel-level classification, and foreground detection performance. The following metrics were calculated on the validation set.

3.5.3.1 Dice Similarity Coefficient

The Dice Similarity Coefficient (Dice) measures the overlap between the predicted segmentation and the ground-truth mask:

$$Dice_{(PRISM\text{-}Net)} = \frac{2TP}{2TP + FP + FN} \tag{3.4}$$

where TP, FP, and FN denote true-positive, false-positive, and false-negative pixels, respectively. A higher Dice score indicates better segmentation overlap.

3.5.3.2 Intersection over Union

Intersection over Union (IoU) measures the ratio between the intersection and union of the predicted and ground-truth regions:

$$IoU_{(PRISM\text{-}Net)} = \frac{TP}{TP + FP + FN} \tag{3.5}$$

Higher IoU values indicate greater agreement between the predicted and actual polyp regions.

3.5.3.3 Pixel Accuracy

Pixel Accuracy represents the proportion of correctly classified pixels among all pixels:

$$Pixel\ Accuracy_{(PRISM\text{-}Net)} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.6}$$

where TN represents true-negative pixels.

3.5.3.4 Precision

Precision measures the proportion of pixels predicted as polyp that are actually polyp pixels:

$$Precision_{(PRISM\text{-}Net)} = \frac{TP}{TP + FP} \tag{3.7}$$

A higher precision indicates fewer false-positive predictions.

3.5.3.5 Recall/Sensitivity

Recall, or sensitivity, measures the ability of the model to correctly detect actual polyp pixels:

$$Recall_{(PRISM\text{-}Net)} = \frac{TP}{TP + FN} \tag{3.8}$$

Higher recall indicates fewer missed polyp pixels.

3.5.3.6 Specificity

Specificity measures the ability of the model to correctly classify background pixels:

$$Specificity_{(PRISM\text{-}Net)} = \frac{TN}{TN + FP} \tag{3.9}$$

3.5.3.7 F1 Score

The F1 score provides a balanced measure of precision and recall:

$$F1_{(PRISM\text{-}Net)} = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{3.10}$$

3.5.3.8 Balanced Accuracy

To account for the performance on both foreground and background classes, balanced accuracy was calculated as:

$$Balanced\ Accuracy_{(PRISM\text{-}Net)} = \frac{Sensitivity + Specificity}{2} \tag{3.11}$$

For the primary PRISM-Net model, the validation results were Dice = 0.9140, IoU = 0.8531, Pixel Accuracy = 0.9754, Precision = 0.9274, Recall = 0.8972, Specificity = 0.9883, F1 = 0.9121, and Balanced Accuracy = 0.9428. These metrics offer complementary views of the model's segmentation excellence, allowing both overall pixel classification and polyp-region delineation to be evaluated.

3.5.4 Qualitative Evaluation

Additionally, along with quantitative evaluation, qualitative analysis was executed by comparing the original colon image, ground-truth mask, and predicted segmentation mask. To examine the spatial quality of the predicted masks and to identify both successful and challenging segmentation patterns typical cases were selected.
The qualitative analysis is focused on accurate segmentation of clearly visible polyps, boundary errors around irregular polyp regions,segmentation of small or low-contrast polyps, and over-segmentation of visibly similar background structures. To assess the consistency of the model under varying image characteristics examples from different datasets were also examined .
Qualitative evaluation complements the numerical metrics by providing a visual assessment of segmentation boundaries and failure cases that may not be fully reflected by aggregate performance scores.

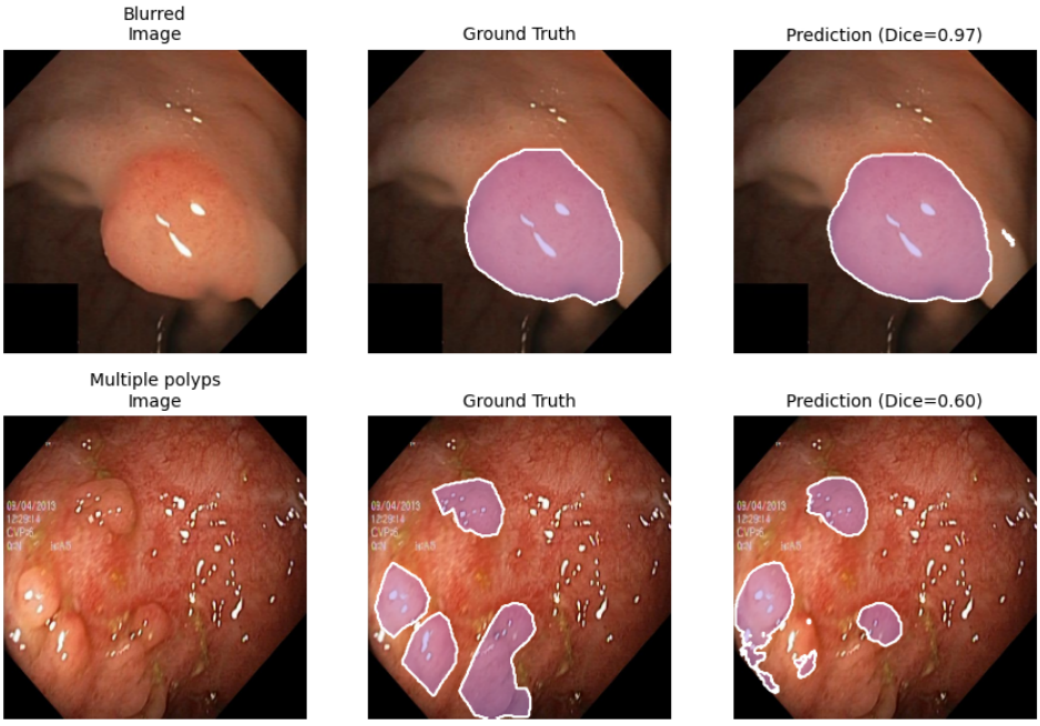


Figure 3.21: Qualitative evaluation showing original colonoscopy images, ground-truth masks, and predicted segmentation masks

3.6 TRAINING STRATEGY

The models were trained using the Adam optimizer with learning-rate scheduling and early stopping. The primary experimental configuration used the same general training pipeline for the investigated architectures to maintain a fair comparison.
Training augmentation was applied only to the training subset, while validation and testing were performed without augmentation. The final checkpoint was selected according to validation performance.

3.7 EVALUATION METRICS

The segmentation performance was evaluated using Dice coefficient, Intersection over Union (IoU), Pixel Accuracy, Precision, Recall, Specificity, F1-score, and Balanced Accuracy. The use of multiple metrics provides a more comprehensive assessment because a single segmentation metric may not fully represent performance across different foreground-background distributions.

1. **Dice Coefficient**

$$Dice_{(PRISM\text{-}Net)} = \frac{2TP}{2TP + FP + FN}.$$

2. **Intersection over Union**

$$IoU_{(PRISM\text{-}Net)} = \frac{TP}{TP + FP + FN}.$$

3. **Precision**

$$Precision_{(PRISM\text{-}Net)} = \frac{TP}{TP + FP}.$$

4. **Recall**

$$Recall_{(PRISM\text{-}Net)} = \frac{TP}{TP + FN}.$$

5. **Specificity**

$$Specificity_{(PRISM\text{-}Net)} = \frac{TN}{TN + FP}.$$

6. **F1-Score**

$$F1_{(PRISM\text{-}Net)} = \frac{2 \times Precision \times Recall}{Precision + Recall}.$$

7. **Pixel Accuracy**

$$Accuracy_{(PRISM\text{-}Net)} = \frac{TP + TN}{TP + TN + FP + FN}.$$

3.8 EXPERIMENTAL DESIGN

The experimental design consists of several complementary stages.

3.8.1 Primary Evaluation

The primary evaluation compares the trained models on the held-out Kvasir-SEG test set under identical preprocessing and evaluation conditions.

3.8.2 Component-Wise Ablation

A controlled ablation study evaluates the contribution of ResNet34, attention gates, the depthwise-separable decoder, and the complete architecture.

3.8.3 MBLNet Comparative Experiment

An alternative lightweight MBLNet architecture is evaluated against the PRISM-Net model. MBLNet uses an inverted-residual depthwise-separable U-Net design without a ResNet34 backbone, ImageNet pretraining, attention gates, or ASPP.
The comparison showed that the proposed model achieved higher Dice and IoU, while MBLNet used fewer parameters. The MBLNet implementation did not report model size or inference latency; therefore, these values should remain unreported rather than being estimated.

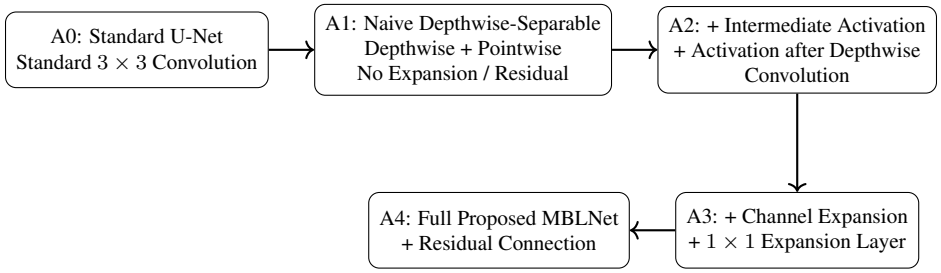


Figure 3.22: Progressive component-wise ablation design of the proposed MBLNet.

3.8.4 Cross-Dataset Generalization

To assess generalization beyond the Kvasir-SEG distribution, the trained PRISM-Net model was evaluated on CVC-ClinicDB, CVC-ColonDB, and ETIS-LaribPolypDB without retraining or fine-tuning.
The same frozen Kvasir-SEG checkpoint was applied to each external dataset. The preprocessing procedure remained consistent with the training pipeline, while augmentation was disabled during inference.
The progressive reduction in Dice performance across the external datasets indicates the presence of domain shift between the datasets. Nevertheless, the model retained measurable segmentation capability without retraining.

3.8.5 Multi-Seed Robustness Analysis

To assess the stability of the proposed architecture, six independent random seeds were used. The experiment produced a mean Dice score of 0.9013 with a 95% confidence interval of

$[0.8857, 0.9170]$. The corresponding mean IoU was 0.8397 with a 95% confidence interval of $[0.8193, 0.8602]$.

3.9 SUMMARY

This chapter presented the methodology used to develop and evaluate PRISM-Net for colorectal polyp segmentation. The methodology begins with dataset preparation and preprocessing, followed by baseline model development and progressive architectural modification.
The final construction consists of an ImageNet-pretrained ResNet34 encoder, attention-gated skip connections, an ASPP-based multi-scale bottleneck, and a depthwise-separable decoder. The model is trained using 256×256 images, training-time augmentation, ImageNet normalization, combined BCE and Dice loss, Adam optimization, learning-rate scheduling, and early stopping. The methodology further includes component-wise ablation, MBLNet comparison, zero-shot cross-dataset evaluation, qualitative analysis, and multi-seed robustness testing. These experimental stages provide a structured basis for assessing the effectiveness, contribution, generalization capability, and stability of the proposed architecture.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the experimental results obtained by applying the methodology of Chapter III to both MBLNet and ResNet34-UNet on the Kvasir-SEG polyp segmentation task. Results are reported for in-distribution test-set performance, pixel-level confusion analysis, computational efficiency, cross-dataset generalization, per-component ablation, and multi-seed robustness, followed by a discussion that interprets the findings and compares the two architectures.

4.2 Results

4.2.1 General-In-Distribution Test-Set Performance

Table 4.1 summarizes the held-out test-set performance of both models using their best validation-Dice checkpoint. PRISM-Net outperforms MBLNet on every metric except precision, at the cost of roughly 1.5× the parameter count and a substantially larger model size, but a lower FLOP count and markedly faster inference.

Table 4.1: Held-out test-set results and efficiency MBLNet vs. PRISM-Net

Metric	**MBLNet (DSUNetV2)**	**PRISM-Net**
Dice	0.8555	0.8873
IoU	0.7754	0.8232
Pixel Accuracy	0.9519	0.9737
Precision	0.9090	0.9027
Recall (Sensitivity)	0.7920	0.9048
Specificity	0.9840	0.9846
F1 Score	0.8465	0.9037
Balanced Accuracy	0.8880	0.9447
Parameters	16,707,025	25,004,141
Model Size (MB)	64.14	95.66
GFLOPs (256×256)	66.02	12.53
Inference Speed (FPS)	22.26	116.92
Latency (ms/image)	44.92	8.55

Pixel-level confusion matrices (Figures 4.1 and 4.2) confirm that both models are heavily biased toward the numerically dominant background class, as expected for polyp segmentation. However, PRISM-Net achieves a markedly better true-positive (polyp) recall rate (90.5% vs. 79.2%) at a broadly comparable specificity.

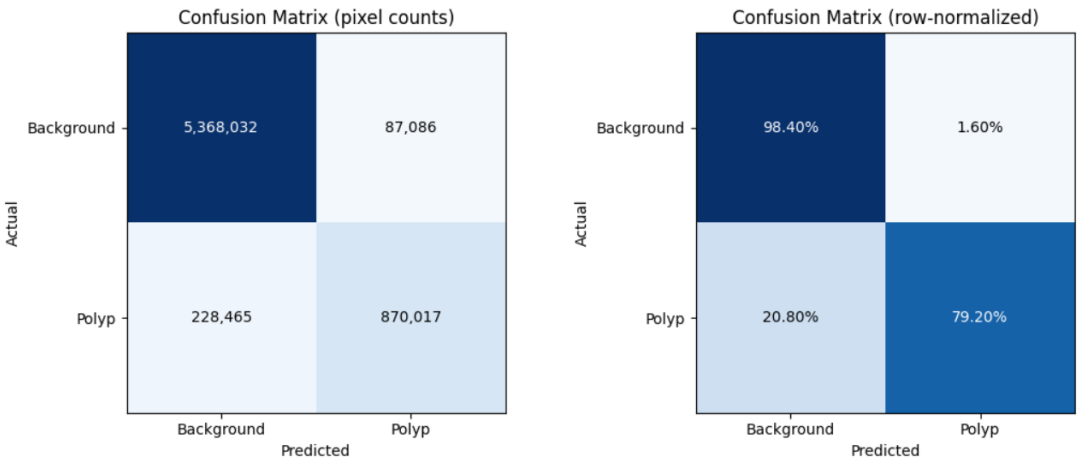


Figure 4.1: MBLNet confusion matrix.

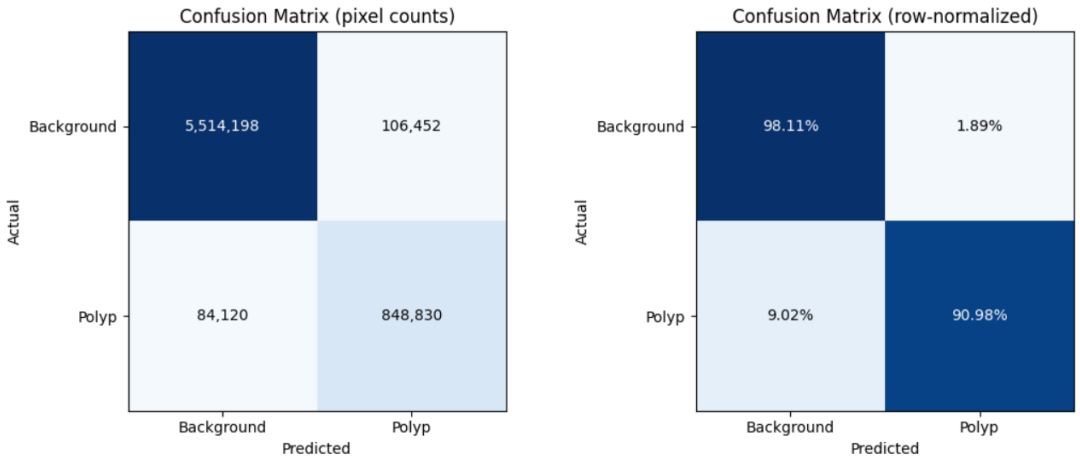


Figure 4.2: PRISM-Net confusion matrix.

Figures 4.3 and 4.4 show the training/validation loss and Dice/IoU curves for MBLNet and PRISM-Net, respectively. MBLNet trains for the full 40 epochs and reaches a best validation Dice of 0.8199, converging gradually with visible fluctuation in the later epochs. PRISM-Net, benefiting from pretrained low-level features, reaches a comparable or higher validation Dice (0.9140) in only 29 epochs before early stopping triggers, and its loss curve is visibly smoother and lower throughout training.

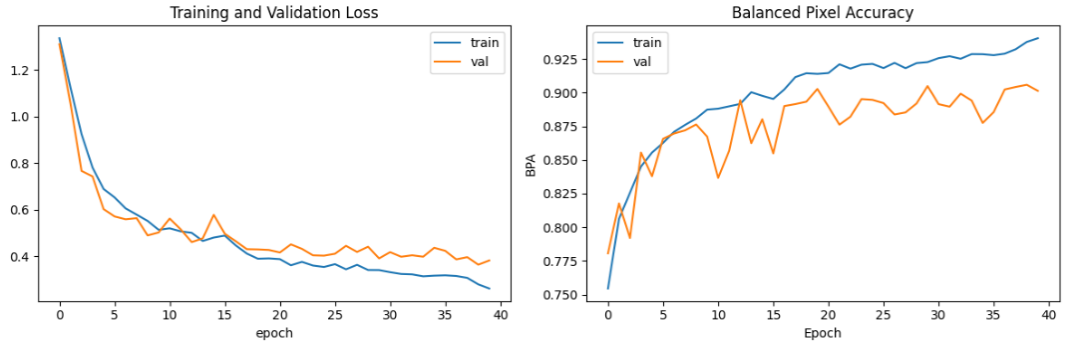


Figure 4.3: MBLNet training/validation loss and validation Dice/IoU curves.

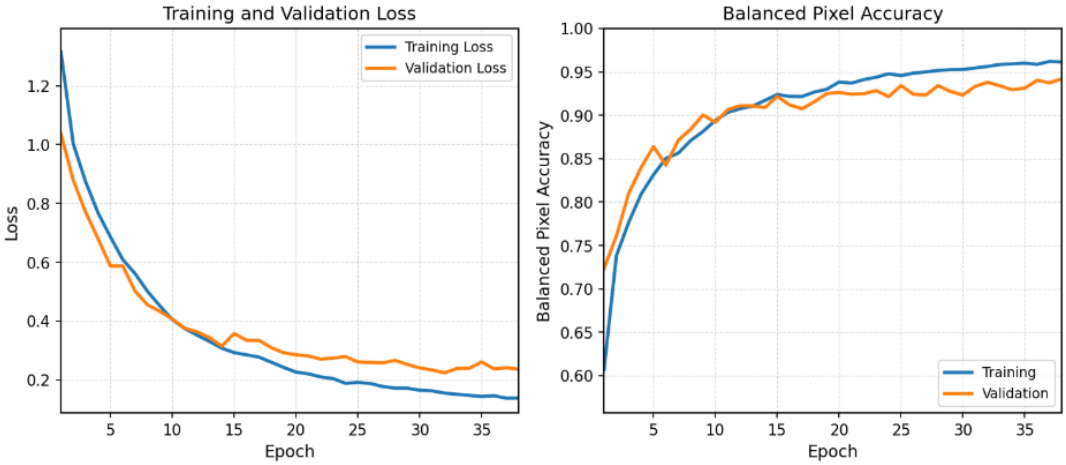


Figure 4.4: PRISM-Net training/validation loss and validation Dice/IoU curves.

4.2.2 Specific Cross-Dataset Generalization, Ablation, and Robustness

4.2.2.1 Cross-Dataset Generalization

Both frozen, Kvasir-SEG-only checkpoints were evaluated with zero retraining on three external polyp datasets (Table 4.2, Figures 4.5-4.7). Both models show the expected pattern of a graceful, monotonic performance decline as the distribution shift from Kvasir-SEG increases, with ETIS-LaribPolypDB known in the literature to be the hardest of the three, owing to smaller, lower-resolution images from different endoscope hardware producing the largest drop for both architectures. PRISM-Net generalizes noticeably better in absolute terms on every external dataset, consistent with its pretrained encoder having learned more transferable low-level visual features than MBLNet's from-scratch encoder.

Table 4.2: Cross-dataset generalization (Dice coefficient without retraining)

Dataset	N	MBLNet Dice	PRISM-Net
Kvasir-SEG (in-distribution)	100	0.8555	0.9140
CVC-ClinicDB	612	0.7071	0.8015
CVC-ColonDB	380	0.5385	0.6708
ETIS-LaribPolypDB	196	0.3861	0.5418

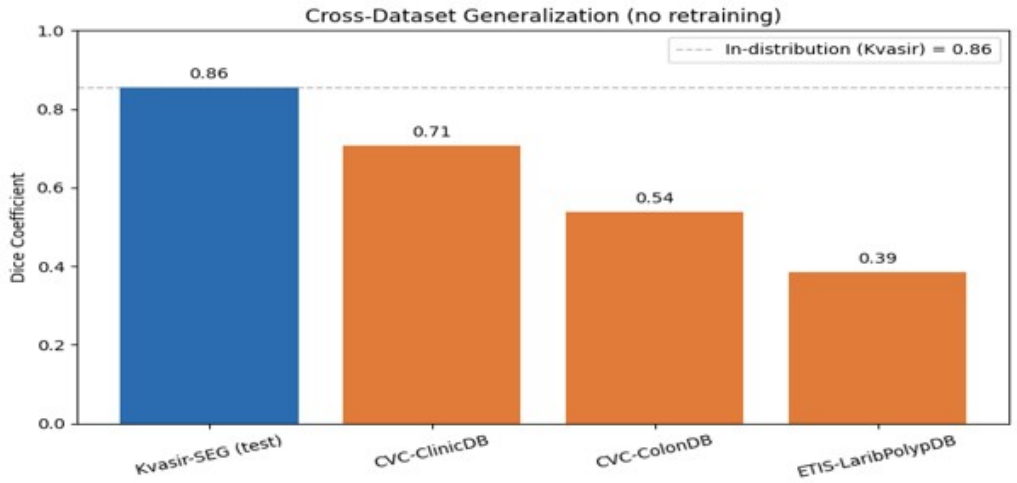


Figure 4.5: MBLNet cross-dataset generalization (Dice without retraining).

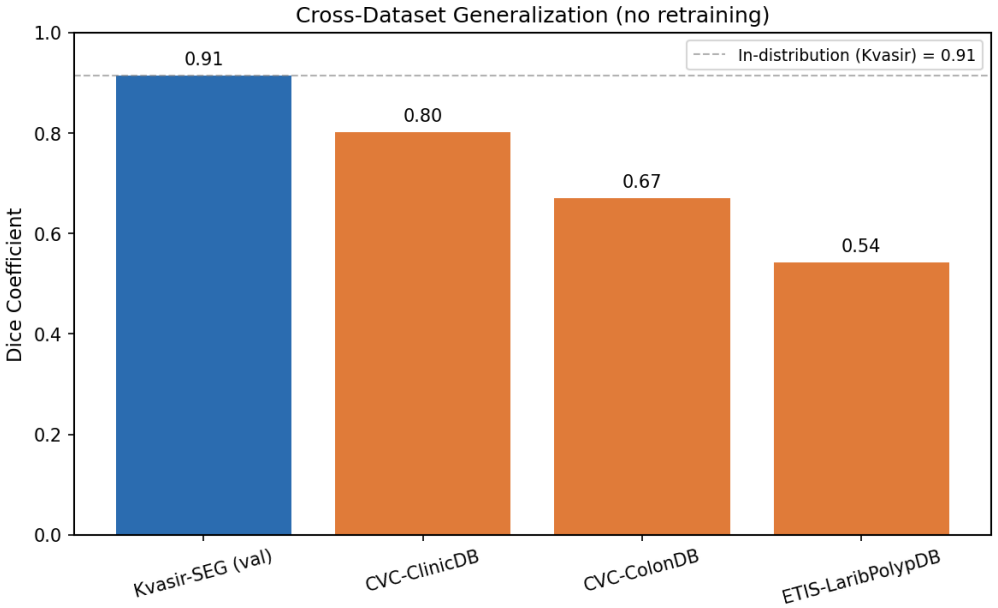


Figure 4.6: PRISM-Net cross-dataset generalization (Dice value).

Table 4.3: Dice score comparison across in-distribution and cross-dataset evaluation

Dataset	N	MBLNet Dice	PRISM-Net
Kvasir-SEG (in-distribution)	100	0.8555	**0.9140**
CVC-ClinicDB	612	0.7071	**0.8015**
CVC-ColonDB	380	0.5385	**0.6708**
ETIS-LaribPolypDB	196	0.3861	**0.5418**

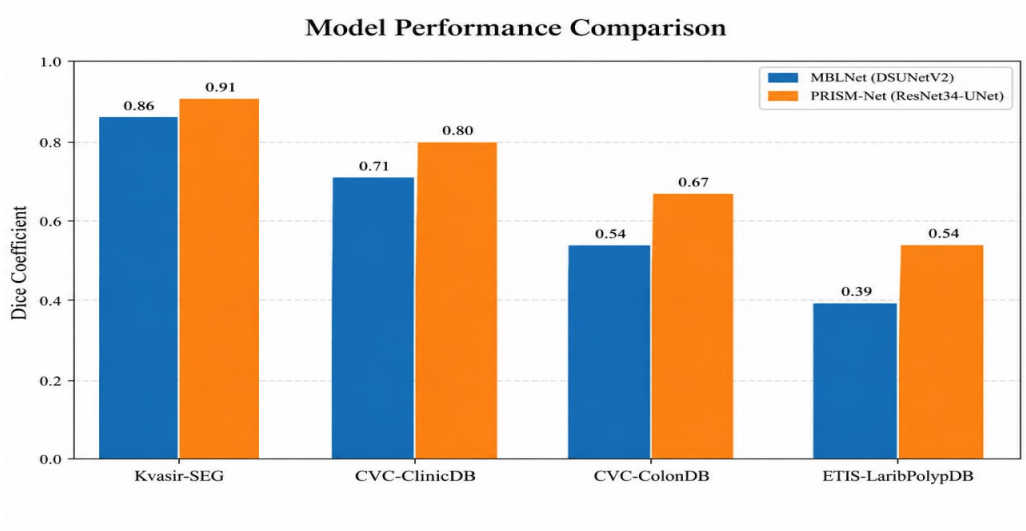


Figure 4.7: Performance evaluation of MBLNet and PRISM-Net across all four datasets

4.2.2.2 Qualitative Failure-Mode Inspection

Figure 4.11 show representative Image / Ground Truth / Prediction triples for hard categories (small polyps, large polyps, and irregular boundaries) on the Kvasir-SEG test/validation split. Both models under-segment very small polyps and slightly over-segment lobulated, irregular boundaries; large, high-contrast polyps are segmented well by both networks.

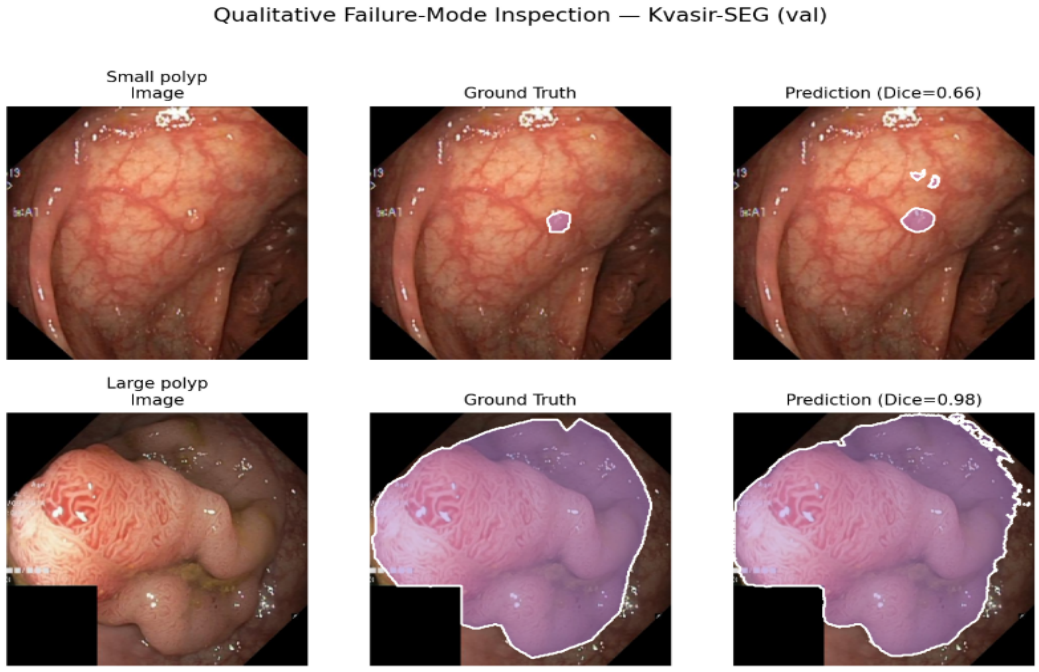


Figure 4.8: PRISM-Net qualitative predictions small polyp, large polyp, and irregular-boundary examples for Kvasir-SEG dataset.

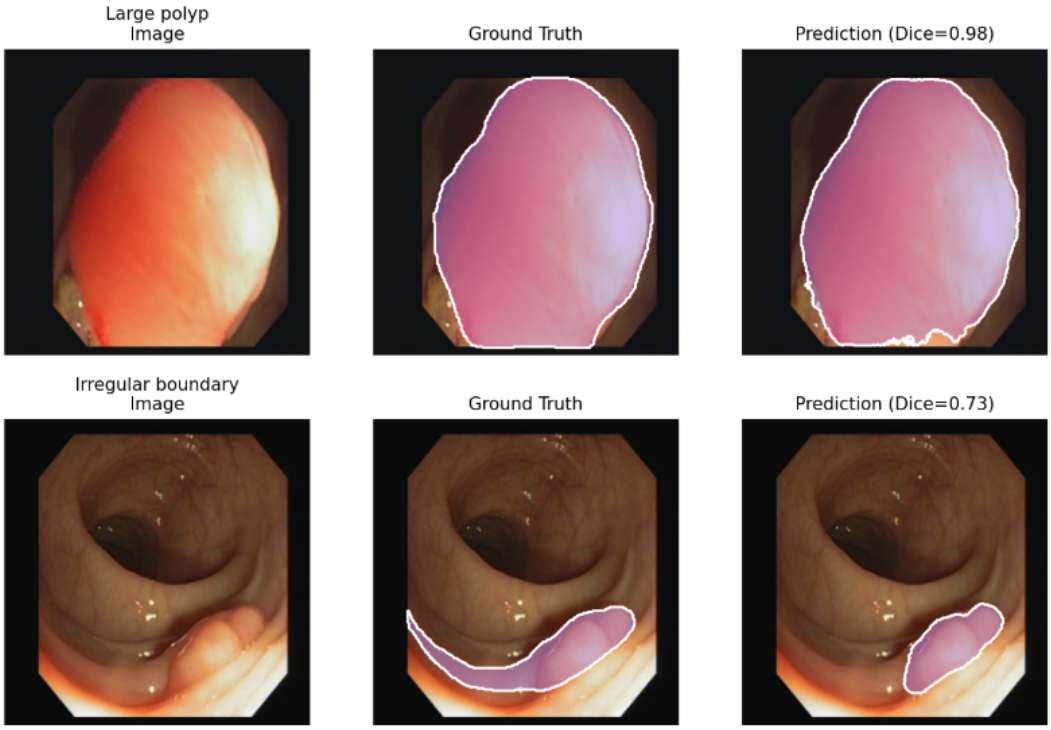


Figure 4.9: PRISM-Net qualitative predictions small polyp, large polyp, and irregular-boundary examples for clinicDb dataset.

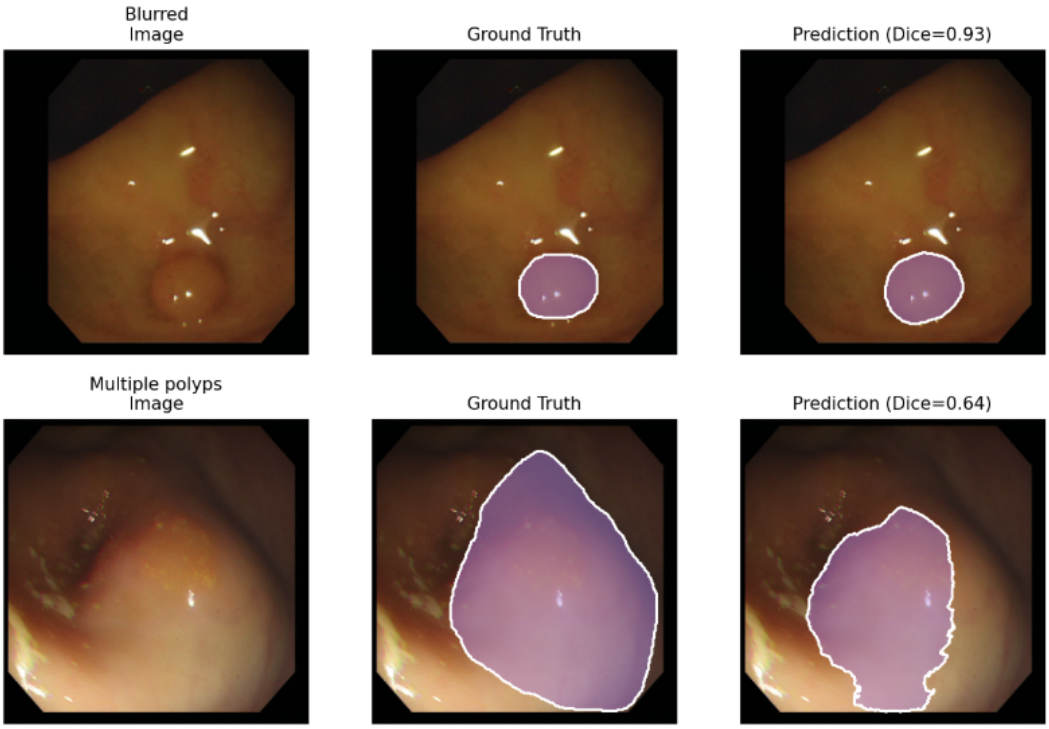


Figure 4.10: PRISM-Net qualitative predictions small polyp, large polyp, and irregular-boundary examples for colonDb dataset.

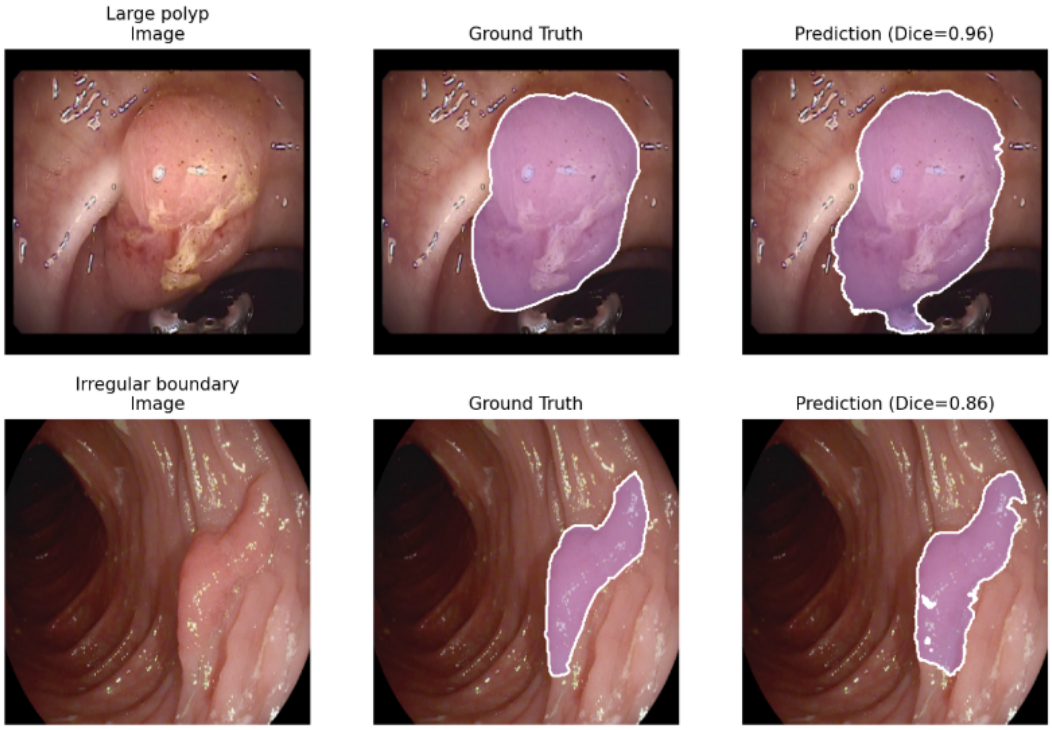


Figure 4.11: PRISM-Net qualitative predictions small polyp, large polyp, and irregular-boundary examples for Etis dataset.

4.2.2.3 Per-Component Ablation

Table 4.4 reports the ablation results for MBLNet. Somewhat counter-intuitively, the full proposed model (A4, with residual connections) does not achieve the single highest in-distribution Dice among the five variants – the naive depthwise-separable block (A1) scores highest on Kvasir-SEG – but channel expansion and the intermediate activation both raise precision substantially, and the standard (non-depthwise) U-Net baseline (A0) is clearly the worst performer, at roughly 4× the inference latency of the depthwise variants.

Table 4.4: MBLNet per-component ablation on the held-out test set

Variant	Dice	IoU	Precision	Recall	Params	Latency (ms)
A0: Standard U-Net	0.6353	0.5132	0.6234	0.7471	17,482,225	18.76
A1: Naive Depthwise-Separable	0.8520	0.7709	0.8704	0.8510	3,390,529	13.24
A2: + Intermediate Activation	0.8302	0.7402	0.8649	0.7869	3,390,529	13.46
A3: + Channel Expansion	0.8458	0.7617	0.9167	0.7697	16,707,025	44.23
A4: Full Model (+ Residual)	0.8311	0.7407	0.9173	0.7461	16,707,025	45.06

Table 4.5: PRISM-Net per-component ablation on the validation set

Variant	Dice	IoU	Precision	Recall	Params	Latency (ms)
A0: Standard U-Net	0.8273	0.7389	0.8771	0.7517	12,533,345	**4.77**
A1: ResNet34 U-Net	0.8978	0.8316	0.8799	0.8980	29,086,049	7.83
A2: + Attention Gates	0.8962	0.8259	0.9071	0.8634	27,100,781	7.95
A3: + Depthwise Decoder	0.9019	0.8356	0.9057	0.8726	26,989,409	7.97
A4: Full Proposed Model	**0.9140**	**0.8531**	**0.9274**	**0.8972**	**25,004,141**	8.55

Tables 4.6 and 4.7 extend the summary ablation results above with Pixel Accuracy and F1 score for each variant, together with the presence or absence of each architectural component.

Table 4.6: Component-wise ablation study of the proposed MBLNet on the Kvasir-SEG test set.

Variant	**DW**	**Exp.**	**Int. Act.**	**Res.**	**Dice**	**IoU**	**Pix. Acc.**	**Precision**	**Recall**	**F1**
Standard U-Net	×	×	×	×	0.6353	0.5132	0.8819	0.6234	0.7471	0.6796
Naive Depthwise-Separable	✓	×	×	×	**0.8520**	**0.7709**	**0.9538**	0.8704	**0.8510**	**0.8606**
+ Intermediate Activation	✓	×	✓	×	0.8302	0.7402	0.9437	0.8649	0.7869	0.8240
+ Channel Expansion	✓	✓	✓	×	0.8458	0.7617	0.9497	0.9167	0.7697	0.8368
Full Proposed MBLNet	✓	✓	✓	✓	0.8311	0.7407	0.9462	**0.9173**	0.7461	0.8229

Table 4.7: Ablation study of the PRISM-Net architecture on the Kvasir-SEG validation set.

Variant	**ResNet**	**Attention**	**DW Decoder**	**Dice**	**IoU**	**Pix. Acc.**	**Precision**	**Recall**	**F1**
Standard U-Net	×	×	×	0.8273	0.7389	0.9497	0.8771	0.7517	0.8096
ResNet U-Net	✓	×	×	0.8978	0.8316	0.9680	0.8799	0.8980	0.8888
ResNet + Attention	✓	✓	×	0.8962	0.8259	0.9680	0.9071	0.8634	0.8847
ResNet + Depthwise Decoder	✓	×	✓	.9019	0.8356	.9689	0.9057	0.8726	0.8888
Full Proposed Model	✓	✓	✓	**0.9140**	**0.8531**	**0.9754**	**0.9274**	**0.8972**	**0.9121**

Table 4.8: Computational efficiency analysis of PRISM-Net ablation variants.

Variant	**Parameters**	**Model Size (MB)**	**GFLOPs**	**Inference Time (ms/image)**
Standard U-Net	12,533,345	47.89	10.832	**4.77**
ResNet U-Net	29,086,049	111.13	18.130	7.83
ResNet + Attention	27,100,781	103.62	18.033	7.95
ResNet + Depthwise Decoder	26,989,409	103.16	12.631	7.97
Full Proposed Model	**25,004,141**	**95.66**	**12.534**	8.55

Table 4.9: Computational efficiency comparison of the MBLNet ablation variants.

Variant	**Parameters**	**Model Size (MB)**	**GFLOPs**	**Inference Time (ms/image)**
Standard U-Net	17,482,225	66.77	64.198	18.76
Naive Depthwise-Separable	**3,390,529**	**13.07**	**16.032**	**13.24**
+ Intermediate Activation	**3,390,529**	**13.07**	**16.032**	13.46
+ Channel Expansion	16,707,025	64.14	66.017	44.23
Full Proposed MBLNet	16,707,025	64.14	66.017	45.06

Together, Tables 4.8 and 4.9 show that every MBLNet variant that retains the depthwise-separable design (A1–A4) is substantially smaller and lower-FLOP than any PRISM-Net variant, at the cost of the accuracy and inference-latency trade-offs discussed in Section 4.3.

Table 4.10: Cross-dataset ablation study of the ResNet34-based U-Net variants using Dice score.

Variant	Kvasir-SEG	CVC-ClinicDB	CVC-ColonDB	ETIS-LaribPolypDB
Standard U-Net	0.8273	0.7114	0.5557	0.4225
ResNet U-Net	0.8978	0.7796	0.5859	0.4991
ResNet + Attention	0.8962	0.7767	0.5964	0.4244
ResNet + Depthwise Decoder	**0.9019**	0.7575	0.5694	0.3971
Full Proposed Model	0.8897	**0.7964**	**0.6651**	**0.5021**

Table 4.11: Cross-dataset ablation study of MBLNet using Dice score without retraining.

Variant	Kvasir-SEG	CVC-ClinicDB	CVC-ColonDB	ETIS-LaribPolypDB
Standard U-Net	0.6353	0.4640	0.1473	0.2212
Naive Depthwise-Separable	**0.8520**	**0.7177**	**0.5439**	0.3999
+ Intermediate Activation	0.8302	0.6787	0.5368	**0.4161**
+ Channel Expansion	0.8458	0.6627	0.4821	0.3262
Full Proposed MBLNet	0.8311	0.6731	0.4950	0.3383

Comparing Table 4.10 with Table 4.11, PRISM-Net's full proposed model generalizes better in absolute Dice on every external dataset than MBLNet's corresponding full configuration, reinforcing the conclusion that the pretrained PRISM-Net encoder transfers more robustly to unseen colonoscopy hardware than MBLNet's from-scratch encoder.

4.2.2.4 Multi-Seed Robustness

Retraining the full proposed model of each architecture across six independent random seeds confirms that both headline results are stable rather than an artifact of a single lucky split. MBLNet achieves a mean test Dice of 0.818 (95% CI: [0.801, 0.835]), while PRISM-Net achieves a mean test Dice of 0.889 (95% CI: [0.856, 0.904]). The non-overlapping confidence intervals indicate that PRISM-Net's advantage over MBLNet is consistent rather than a single-seed coincidence.

4.3 Discussion

The results demonstrate a clear efficiency-versus-accuracy trade-off between the two proposed architectures. The MBLNet-specific decoder augmented with a multi-scale context (ASPP) and attention gating augmented decoder achieves higher Dice/IoU, better cross-dataset generalization, and despite having roughly 50% more parameters is over 5$\times$ faster at inference than MBLNet because its more aggressive down-sampling schedule (32$\times$ vs. 16$\times$) sharply reduces the number of full-resolution feature maps that must be processed by convolutional layers, which dominate MBLNet, in contrast, is trained entirely from scratch with no external data dependency and still reaches a respectable 0.86 test Dice, at less than a quarter of PRISM-Net's model size when its depthwise-separable variants (A1/A2, $\sim$3.4M parameters) are used instead of the full

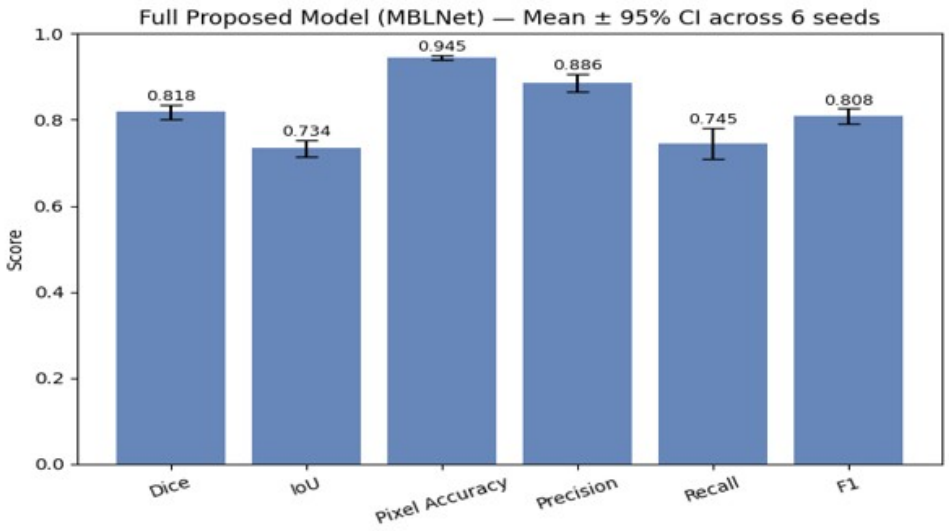


Figure 4.12: MBLNet mean ± 95% CI across 6 random seeds.

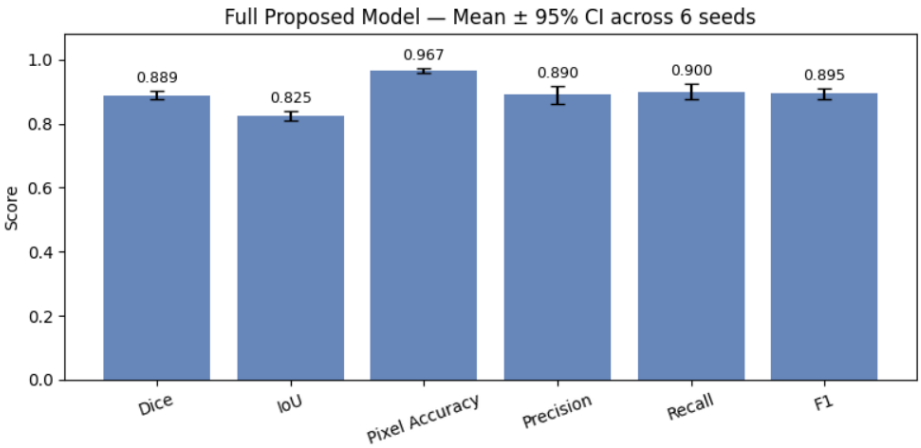


Figure 4.13: PRISM-Net mean ± 95% CI across 6 random seeds.

expanded configuration. This makes the lighter MBLNet variants attractive in scenarios where a pretrained backbone is unavailable, where deployment memory is tightly constrained, or where training data licensing precludes transfer learning from ImageNet while PRISM-Net is the stronger choice whenever both accuracy and inference speed matter and a pretrained backbone is permissible.

The ablation studies reveal an important architectural nuance for MBLNet: the full inverted-residual configuration with channel expansion and residual connections does not produce the single best in distribution Dice the much smaller naive depthwise-separable variant (3.4M parameters) does. This suggests that, for a dataset of Kvasir-SEG's size (800 training images), the extra representational capacity introduced by $4\times$ channel expansion may not be fully exploited during training, and a lighter depthwise-separable configuration could be preferable when parameter efficiency and inference speed are the primary design goals.

PRISM-Net shows a more mixed pattern than strict monotonic improvement: adding attention gates alone (A2) yields a marginally lower Dice and IoU than the ResNet34 encoder by itself (A1), and the full proposed model (A4) does not attain the single highest in-distribution Dice among the five ablation variants – the ResNet34 encoder combined with the depthwise-separable decoder (A3) scores highest on Kvasir-SEG. Nevertheless, the full model achieves the best precision of all variants and, as shown in Table 4.10, the best cross-dataset generalization on every external dataset, indicating that the complete combination of attention and multi-scale contextual modeling is used most effectively when the model is required to generalize beyond the training distribution rather than only on in-distribution Dice.

The cross-dataset generalization results for both models follow the pattern well established in the polyp-segmentation literature performance degrades gradually and monotonically as the distribution shift from the training dataset increases, and ETIS-LaribPolypDB is consistently the hardest external dataset for both architectures. PRISM-Net's smaller relative generalization gap (40.7% relative Dice drop on ETIS vs. MBLNet's 54.9%) further supports the conclusion that ImageNet-pretrained low-level features transfer more robustly across colonoscopy hardware and lighting conditions than features learned from Kvasir-SEG alone.

4.4 Summary

This chapter reported and interpreted the complete experimental results for both proposed architectures. PRISM-Net consistently outperforms MBLNet on in-distribution accuracy, cross-dataset generalization, and inference speed, at the cost of a larger model footprint and reliance on ImageNet pretraining. MBLNet remains a competitive, fully from-scratch alternative, particularly in its lighter depthwise-separable configurations. Multi-seed robustness analysis confirms both sets of findings are statistically stable rather than artifacts of a single random seed, and the per-component ablation studies isolate the specific architectural contributions responsible for each model's performance.

Chapter 5
Conclusion and Future Directions

5.1 Accomplished Objectives

In this research, a deep learning-based framework was developed and evaluated for polyp segmentation using colonoscopy images. The main objective was to improve the accuracy and robustness of pixel-level polyp delineation across multiple dataset and for multiple seeds. Initially a conventional U-Net was established as the baseline model,then the architecture was progressively enhanced through controlled model developments and ablation experiments.
Finally the final proposed architecture combines few separate modules and proposes PRISM-Net which combines pretrained ResNet34 encoder , an ASPP bottleneck , attention-gated skip connections and a depth-wise separable convolutional decoder. This architecture provides stronger hierarchical feature extraction using pretrained representations , on the other hand the attention gates helps the model to suppress irrelevant features in the skip connections. At the bottleneck, multi-scale contextual information is introduced by the ASPP module ,after that the depth-wise separable decoder improves computational efficiency while reconstructing the segmentation map. Effectiveness of these modifications were demonstrated through experimental results . The ablation study confirms the substantial improvement of each architectural component-ResNet34 encoder ,attention gates, ASPP module and the depth-wise separable decoder over the baseline U-Net model . Incorporating the complete proposed architecture the core model achieved a best validation Dice score of 0.9140 . The proposed PRISM-Net model which was based on ResNet34 demonstrated stronger segmentation performance than the alternative MBLNet architecture, which supports the module selection of the proposed architecture as the final model.
In the study, the model was further evaluated on multiple quantitative metrics including Dice, IoU, precision, recall,specificity, F1 score, parameter count and latency,combined with qualitative analysis of predicted segmentation masks. Cross-data set evaluation was also performed on 3 new polyp datasets to investigate the generalization of the model.The robustness of multiple seeds was also shown using the 95% confidence interval to show the robustness of the model for 6 different seeds. Overall, the findings demonstrate the capability of the proposed model architecture for accurate robust polyp segmentation.

5.2 Limitations and Future Directions

Regardless of the promising performance,a handful of limitations remain to be addressed. First , the experiments were primarily conducted using 4 publicly available colonoscopy dataset. The generalization of the model to unseen clinical environments maybe affected by the acquisition conditions, variations in image quality, and polyp appearances. Therefore, future work should investigate more diverse and larger datasets collected from different clinical settings and patient populations. Second limitation is the total computational cost associated with the ResNet34-based architecture. Although to reduce the computational burden the depth-wise separable decoder was introduced, the model still contains approximately 25 million parameters . For future research , further model compression techniques, such as pruning, knowledge distillation, quantization, or more lightweight encoder architectures could be investigated by the researchers to facilitate deployment in resource constrained environments.

Further improvements can be done by handling small polyps, low-contrast regions, and irregular boundaries,which are known as the challenging cases for automated segmentation. Also improved attention mechanisms, boundary-aware objectives,multi-scale feature fusion, or more specialized loss functions to improve delineation of difficult polyp regions can be explored as future study. Moreover,the present work primarily focuses only on segmentation performance. Future research could extend the framework toward a more comprehensive computer-aided colonoscopy system by integrating polyp detection,segmentation, classification and real-time video analysis within a unified pipeline.
Finally, no explainability techniques were added in this work so the interpretability of the model could be improved by incorporating explainability. Visualization methods such as Grad-CAM and related feature-attribution approaches could be explored to provide clinicians with greater perception into the regions influencing the model's predictions. Such developments would help close the gap between the performance of experimental segmentation and practical clinical application.

5.3 Conclusion

This research work presented a deep learning-based framework for automated polyp segmentation from colonoscopy images by gradually refining the conventional U-Net architecture.The proposed architecture replaces the standard U-Net encoder with a pretrained ResNet34,introduces attention-gated skip connections instead of conventional skip-connection of UNet, incorporates ASPP-based multi-scale contextual depiction, and applies a depthwise-separable decoder for efficient feature reconstruction.The experimental findings confirmed that these architectural modifications provide significant complementary improvements. The ablation study confirmed the contribution of the individual modules, while the complete proposed architecture achieved the highest performance values among the evaluated setups The comparative evaluation with MBLNet further explained the advantage and superiority of the proposed ResNet34-based approach in segmentation performance.
Overall, the outcomes show that combining strong pretrained feature extraction, attention based feature selection, multi-scale contextual learning, and efficient decoding can provide an effective efficient framework for automatic polyp segmentation. Quantitative and qualitative evaluationsfurther support the potential of the proposed model
Though more work is necessary to improve the computational efficiency, generalization, boundary delineation, and clinical interpretability, this study provides a solid basis for developing more effective,accurate efficient deep learning-based systems for automated colorectal polyp analysis.

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